

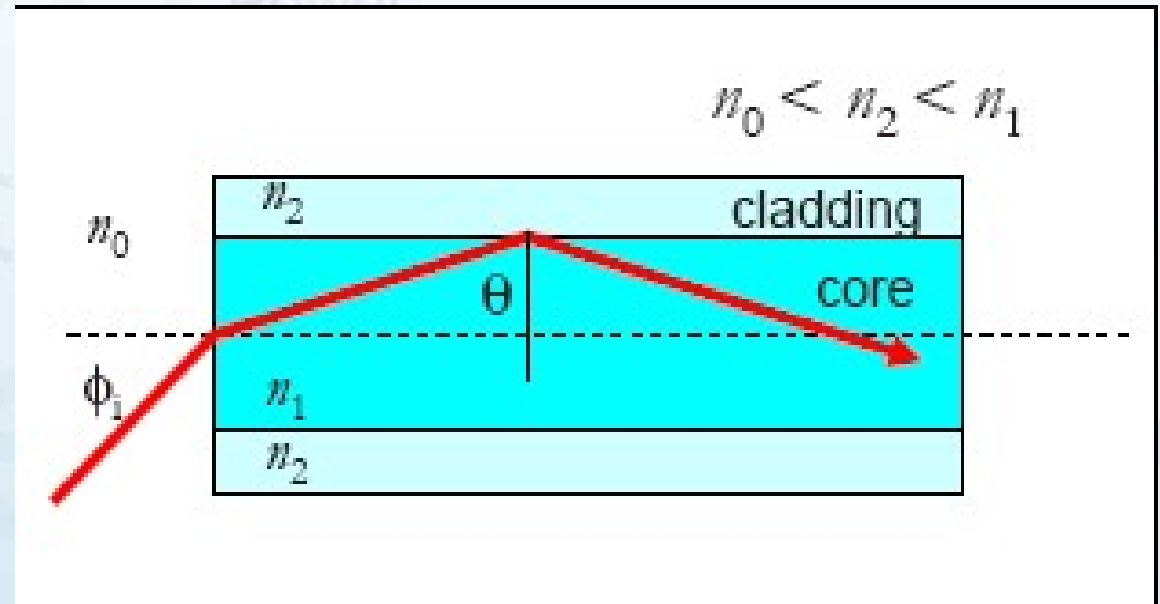
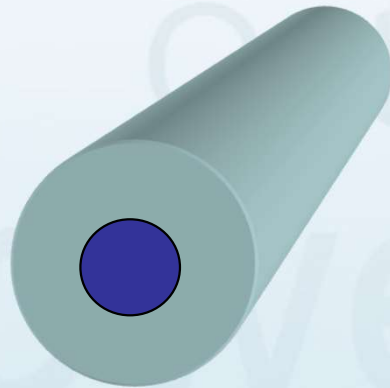
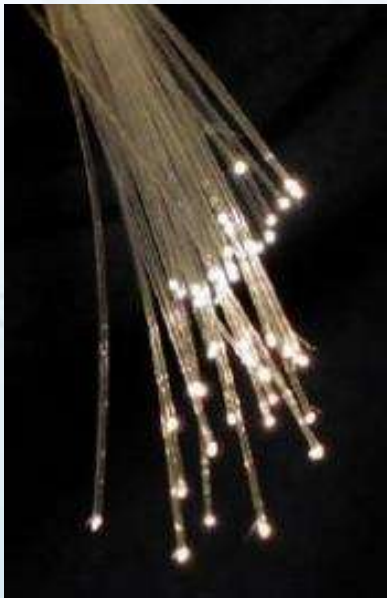
Optical Fiber Technology

Perry Shum

Introduction

Conventional optical fiber

modify



Guiding mechanism: TIR (Total Internal Reflection)

Microstructured Optical Fiber (MOF)

An optical fiber with its feature size in micrometer or sub-micrometer scale

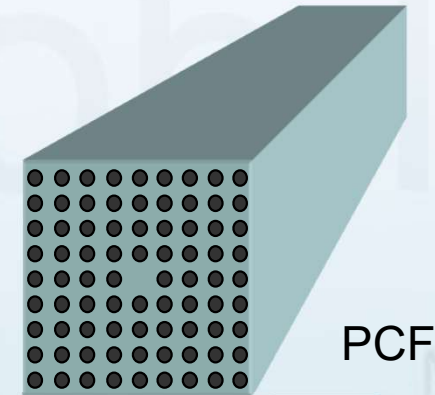
MOF

PCF (Photonic Crystal Fiber)

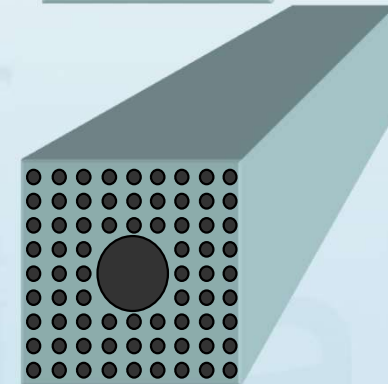
HF (Holey Fiber)

Photonic Bandgap Fiber

Bragg Fiber



PCF



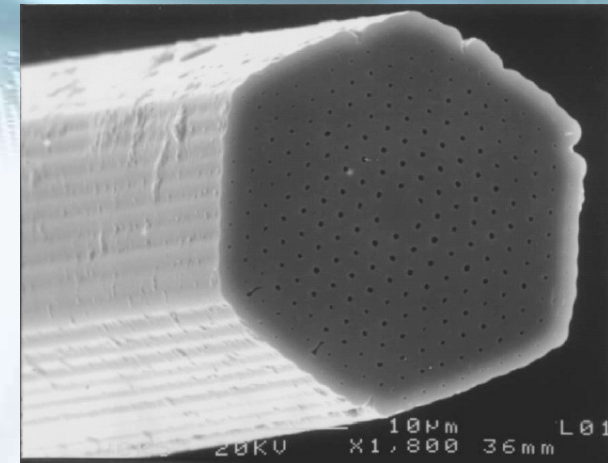
Photonic Bandgap Fiber

1. Guiding mechanism

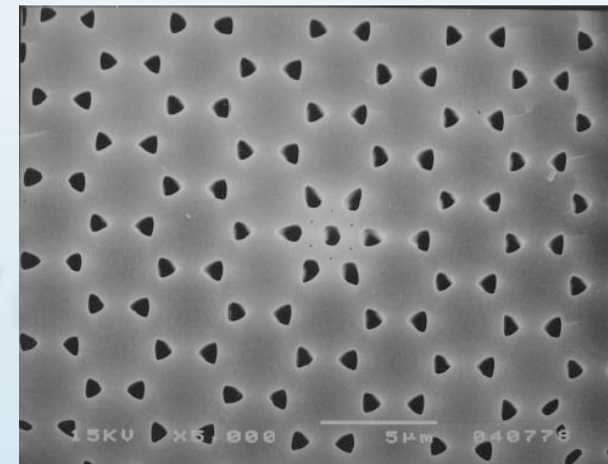
- ❑ Total Internal Reflection
- ❑ Photonic band gap



Pioneered by UK researchers in the 1990's



First index-guiding MOF¹



First photonic bandgap fiber²

[1] Opt. Lett. **21**, 1547 (1996).

[2] Science **282**, 1476 (1998).

3. Interesting properties of MOFs

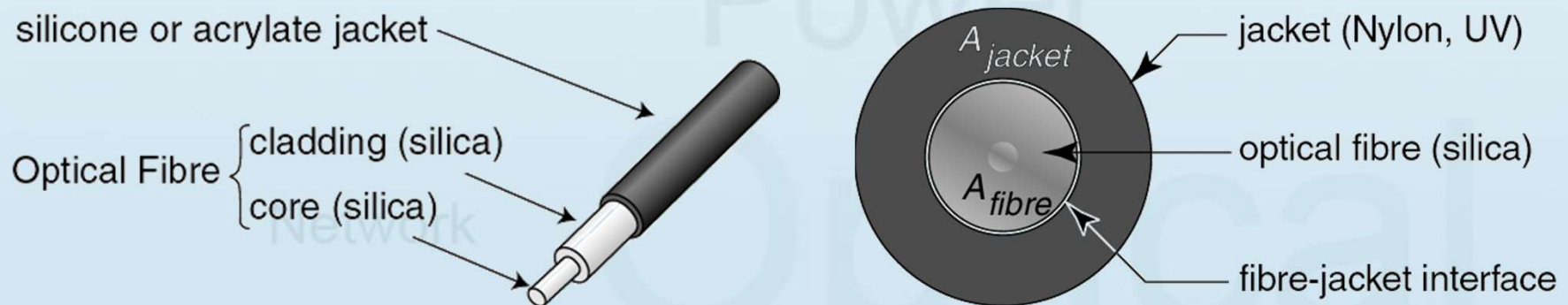
- ❑ Very large numerical aperture.
- ❑ Endlessly single-mode.
- ❑ Low bend loss.
- ❑ Extremely small or large nonlinearities.
- ❑ Air guidance.
- ❑ The possibility to fill gases or liquids into the holes.
- ❑ High birefringence.
- ❑ Dispersion tailoring.

4. Applications

- ❑ Telecom components (dispersion control, polarization control, filtering or switching).
- ❑ High-power fiber lasers and amplifiers.
- ❑ High power delivery.
- ❑ Nonlinear fiber optics.
- ❑ Sensor applications

Optical fiber sensor systems

- Optical fiber as a sensing element
 - Measurement of the modification of the propagation path (delay, attenuation, polarization,...)
 - Measurement of the interaction of light with the propagation medium



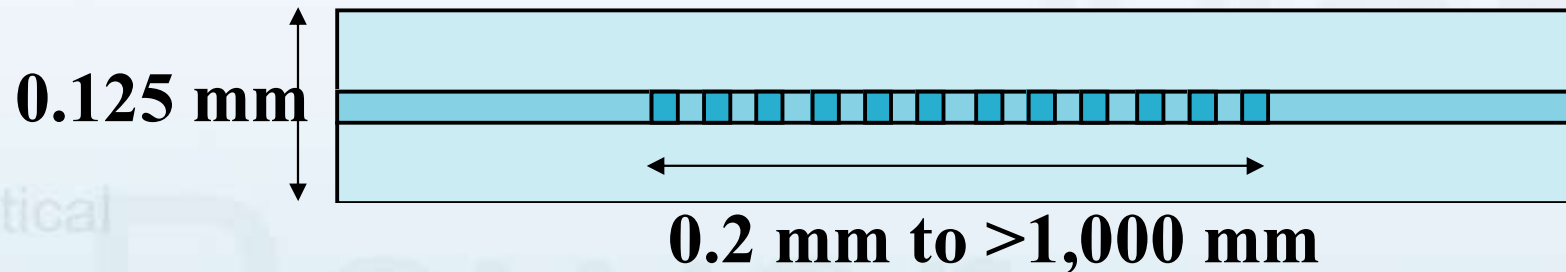
Optical fiber sensor systems

Advantages of Optical Fiber Sensors

- Facilitates distributed sensing systems
- Capability to multiplex large number of sensors > 100 's
- Remote sensing > 50 km measurement range
- Immunity to EMI
- High accuracy - FOG exhibits stability of 1 part in 10^{18}
- Explosion proof
- Small size & light weight
- Self-referencing - no recalibration/re-initialization

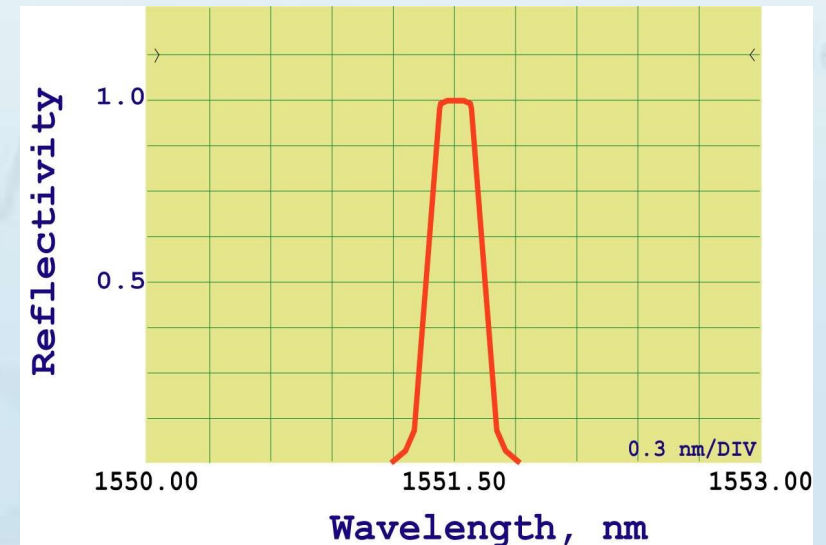
Fiber Bragg Grating-based Sensors

- FBGs are all-silica, glass fiber with grating written inside its 10- μm fiber core



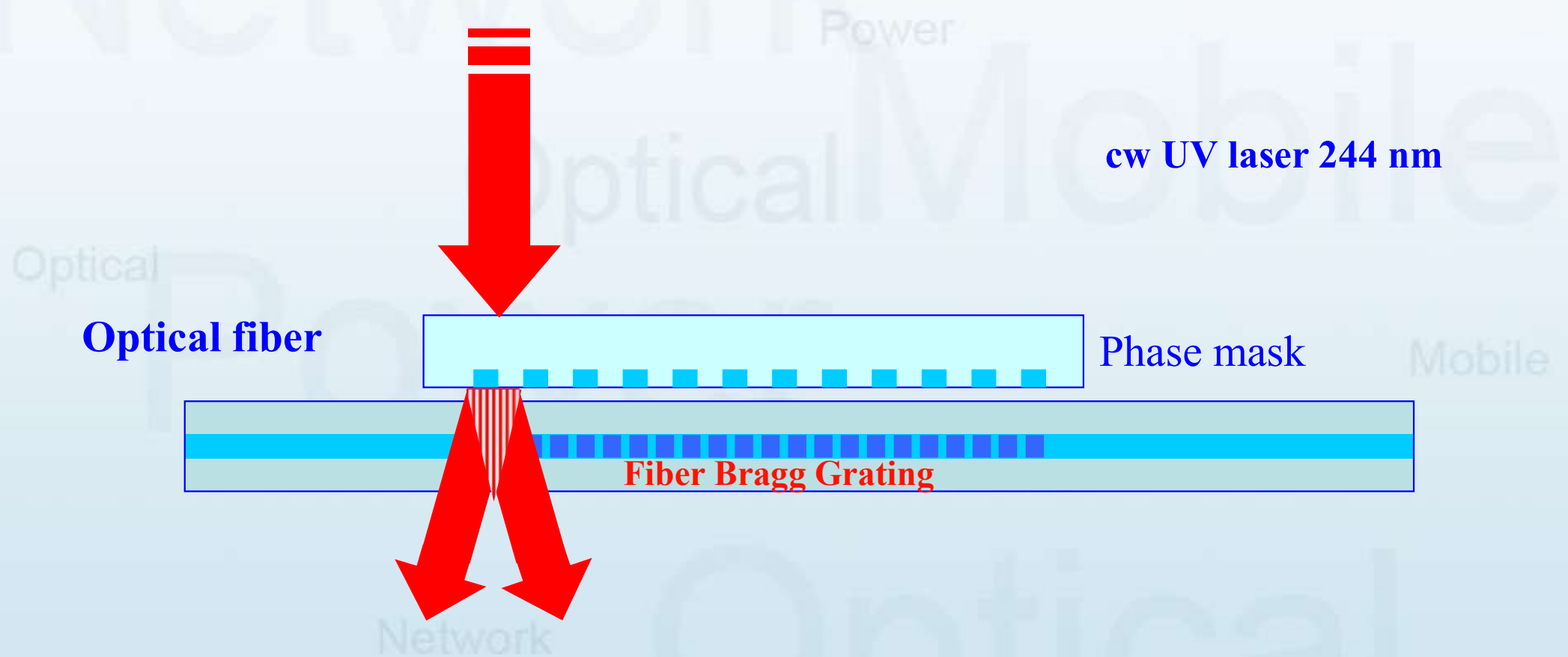
Bragg Condition: $\lambda_B = 2n_{\text{eff}}\Lambda$

- Typical strain response $\sim 1 \text{ pm}/\mu\epsilon$
- Typical temperature response $\sim 10 \text{ pm}/^\circ\text{C}$



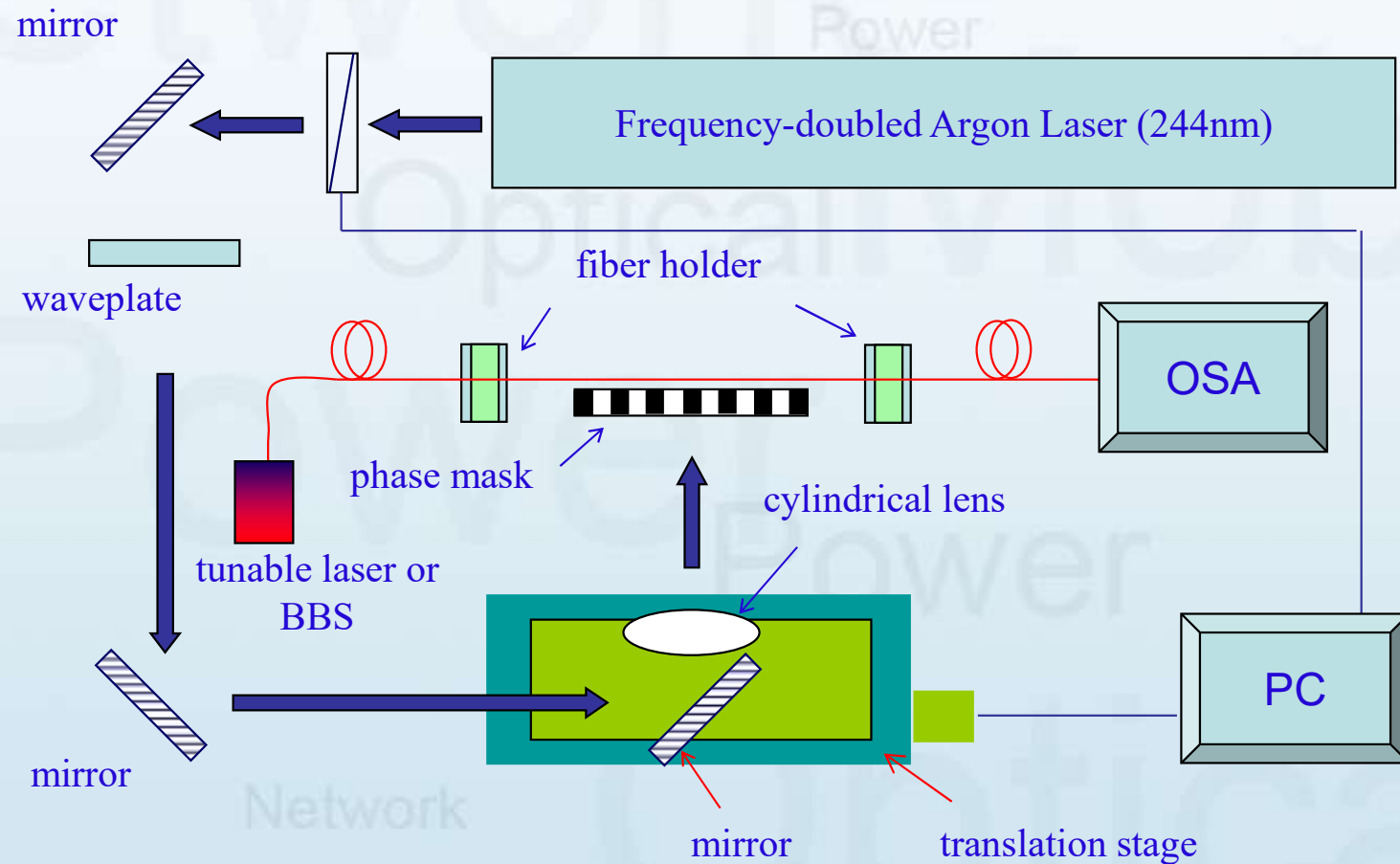
Fiber Bragg Grating-based Sensors

Fabrication of Fiber Gratings



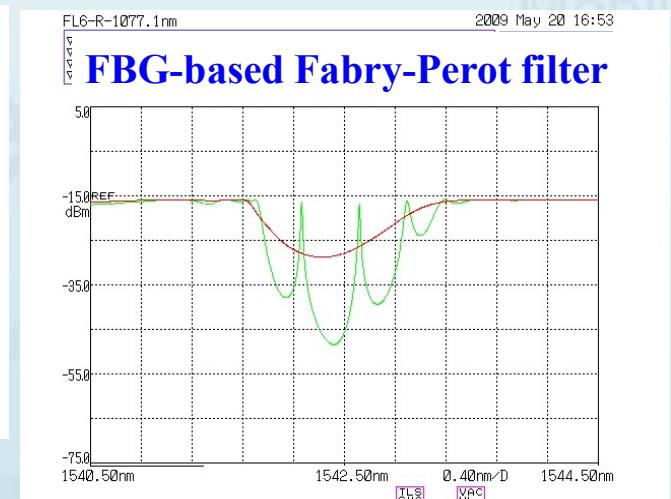
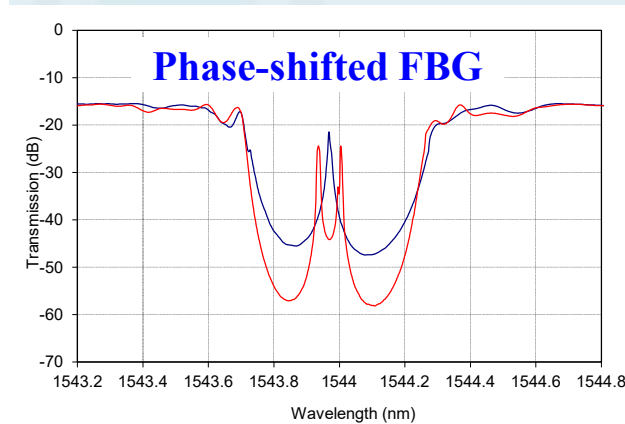
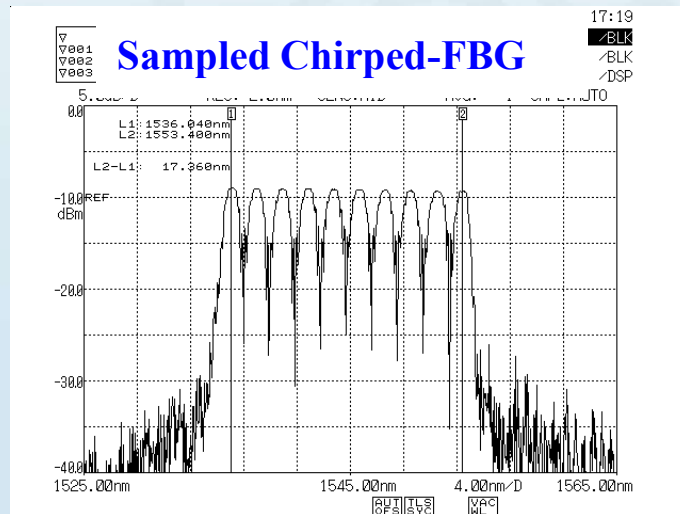
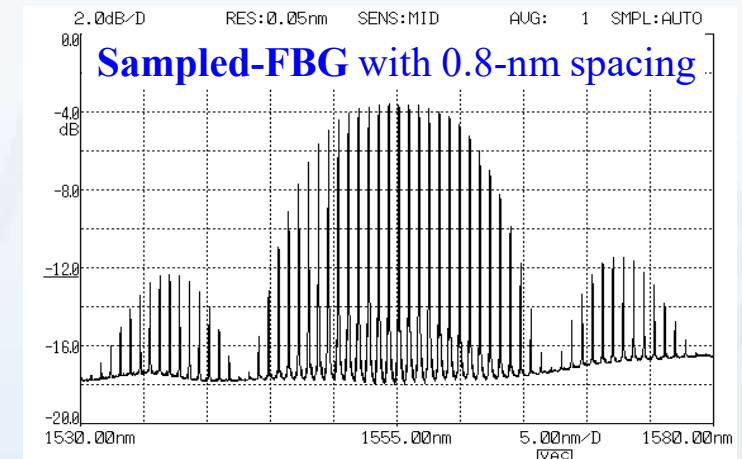
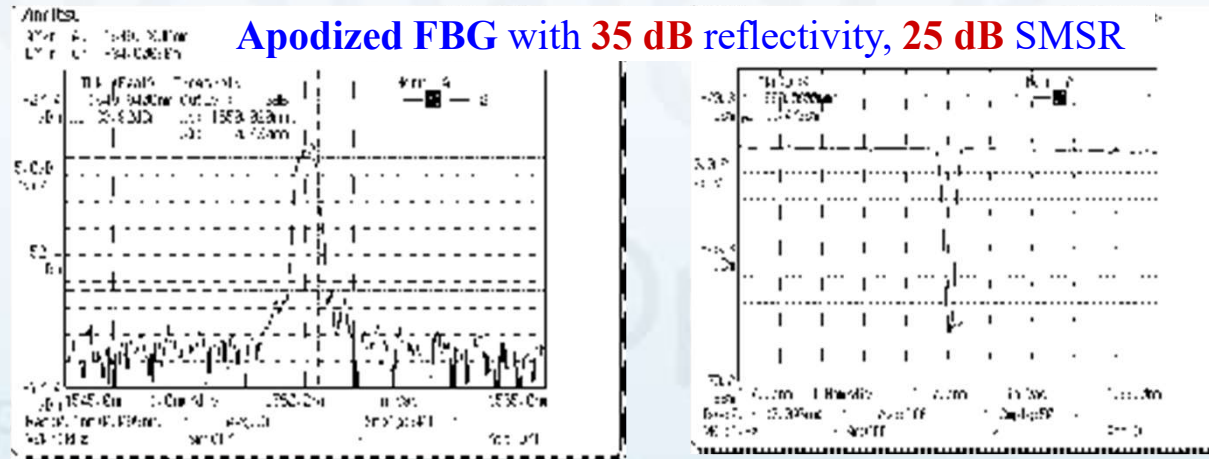
Fiber Bragg Grating-based Sensors

Fabrication of Fiber Gratings



Fiber Bragg Grating-based Sensors

Fabrication of Fiber Gratings

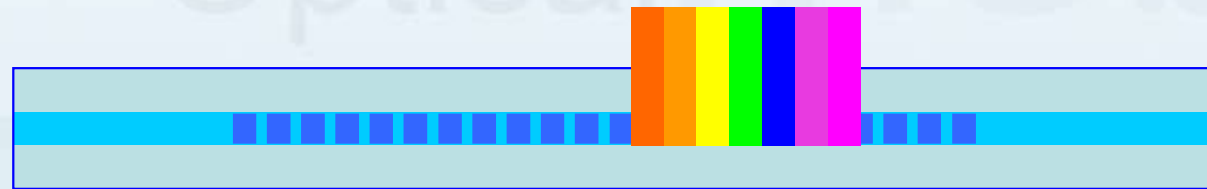


Fiber Bragg Grating-based Sensors

Interrogation of Fiber Bragg Grating

Illuminates FBG with
a broadband light source

The rest of the spectrum is transmitted



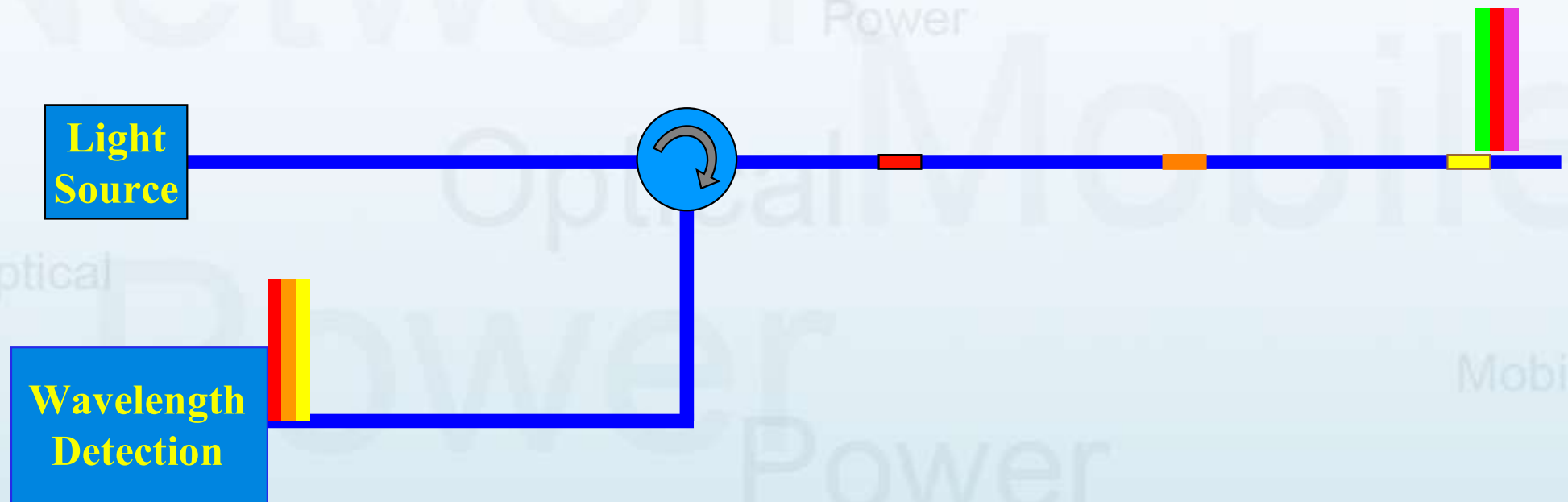
Fiber Bragg Grating

Optical fibre

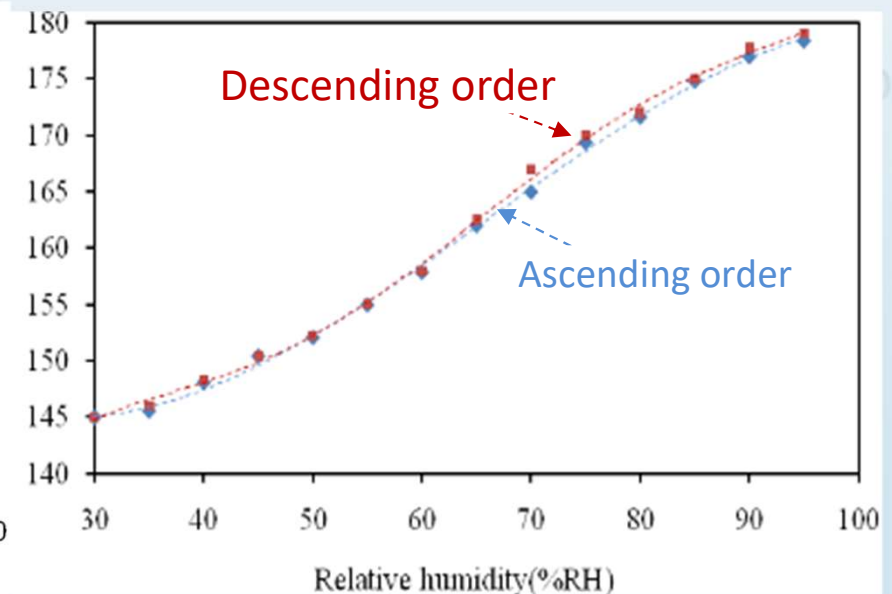
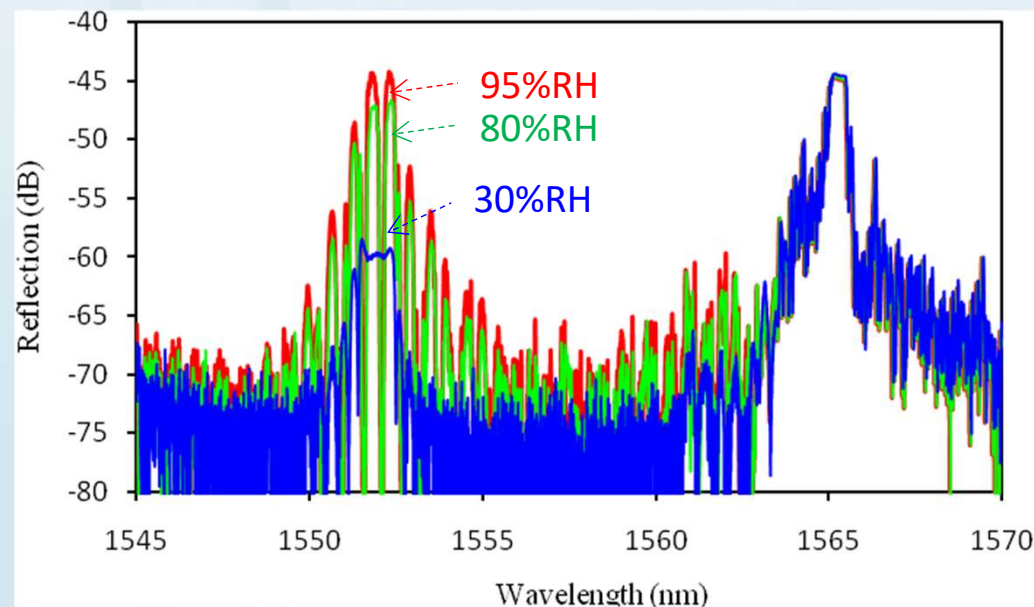
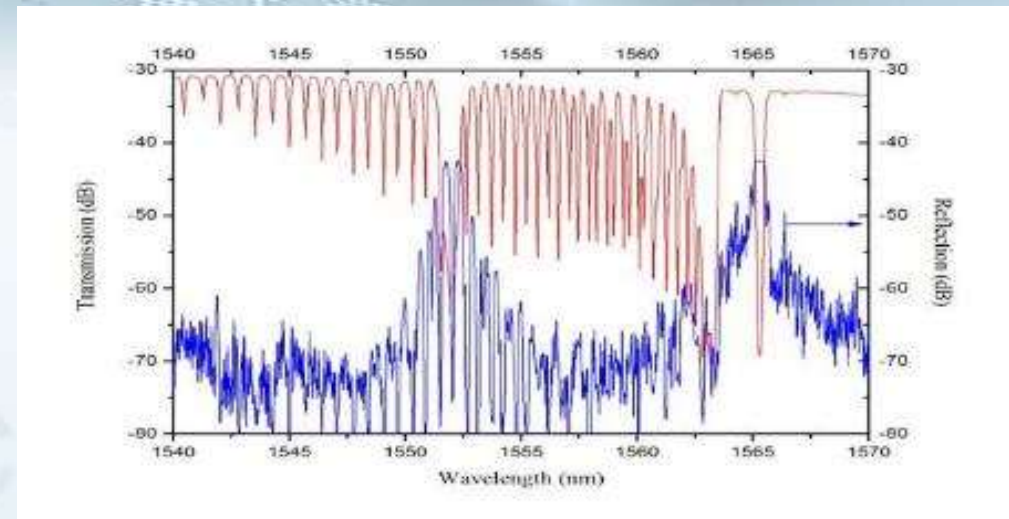
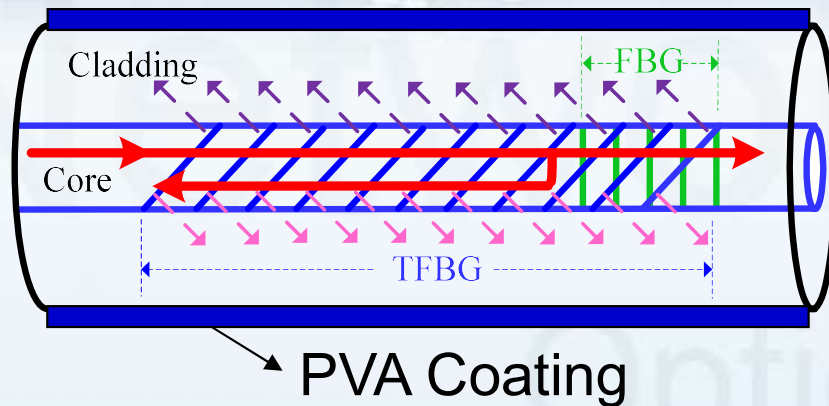
A narrowband of light
reflected by the FBG

Fiber Bragg Grating-based Sensors

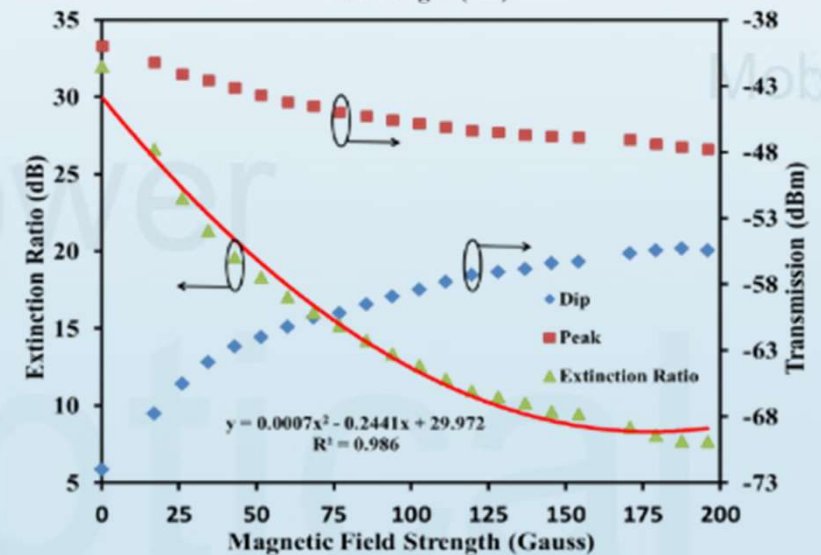
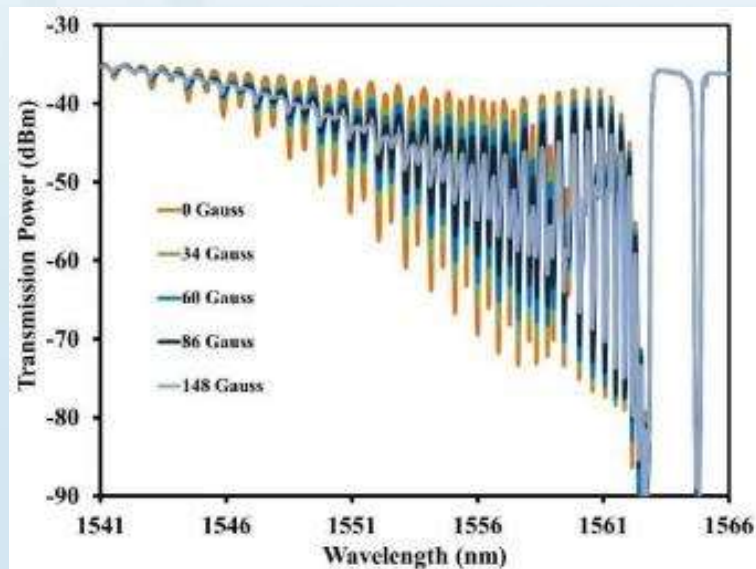
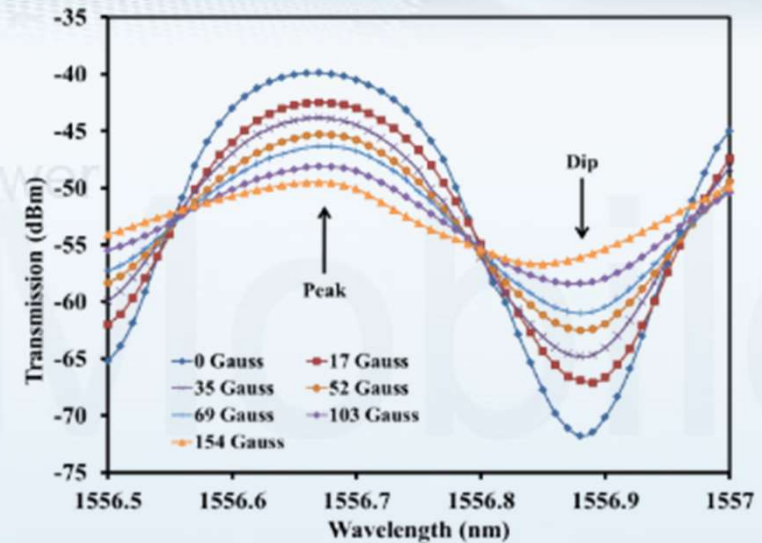
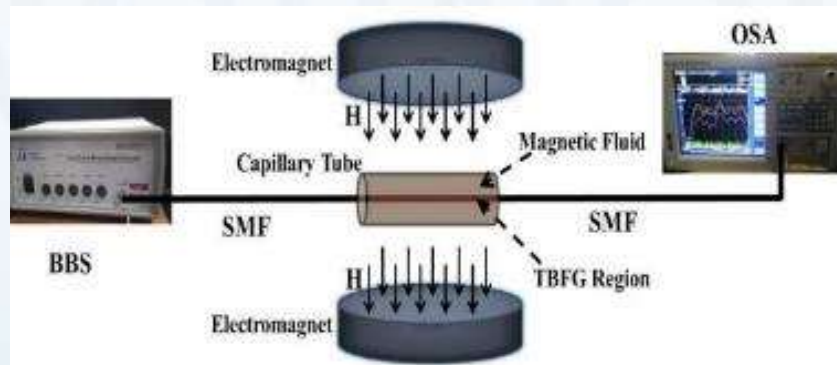
Wavelength-Division Multiplexing of FBG Sensors



Tilted-FBG based humidity sensor



TFBG-based Magneto-optic Field Sensors

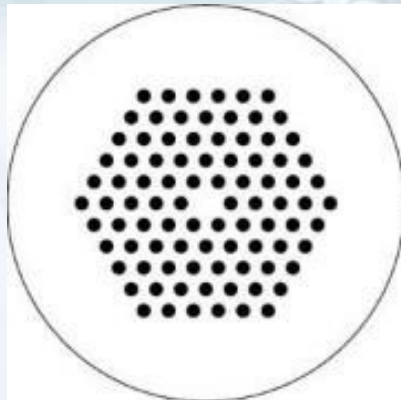


MOF

Design and modeling of MOFs

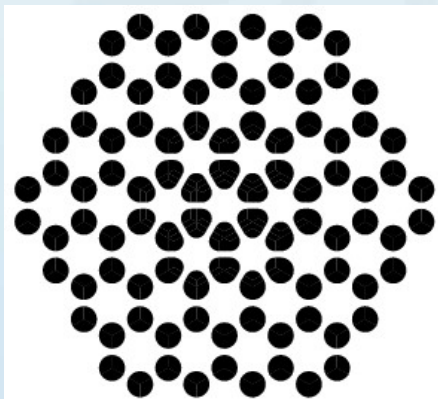
1. Generalized waveguiding principle
2. Liquid crystal filled photonic crystal fiber
3. Improved triangular-lattice photonic crystal fiber
4. Air-guiding honeycomb bandgap fiber
5. Nano-fibers
6. Terahertz kagome-lattice hollow core fibres
7. Compact two-dimensional finite-difference time-domain (C2D-FDTD) method
8. 3D FDTD model with moving problem space
9. Hole-assisted lightguide fibers
10. Dispersion-flattened PMPCF
11. PCFs with high and flattened dispersion
12. Broadband hollow-core Bragg fiber
13. Side-channel PCF

1. Generalized waveguiding principle

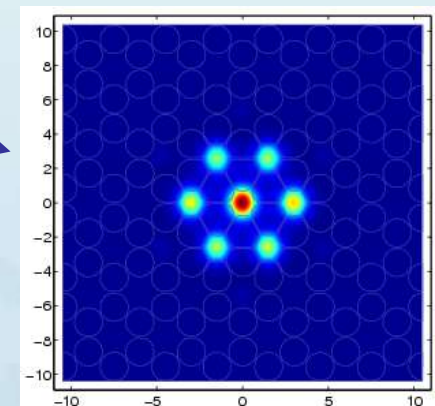
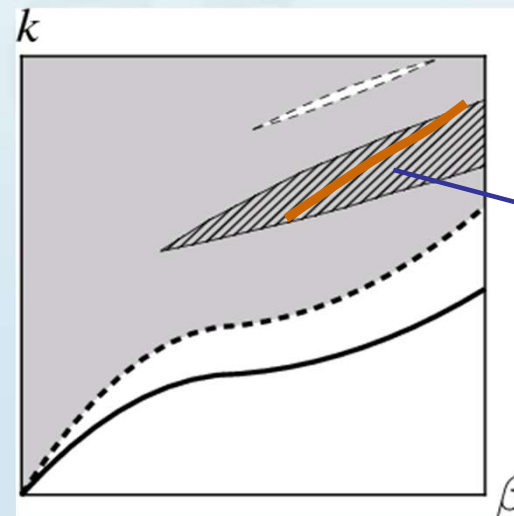
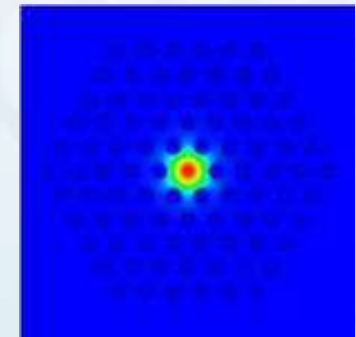
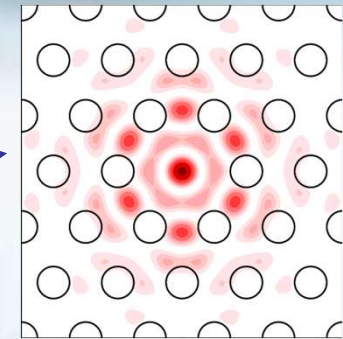
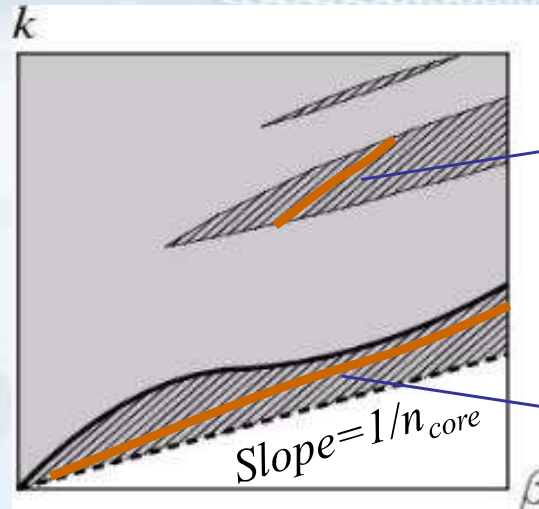


□ silica
■ air

Index-[?]guiding PCF



Heterostructured MOFs



M. Yan, P. Shum, IEEE Photon. Technol. Lett. 1438, 2005.

M. Yan and P. Shum, J. Opt. Soc. Am. B, 1684, 2006

2. Liquid crystal MOF: background

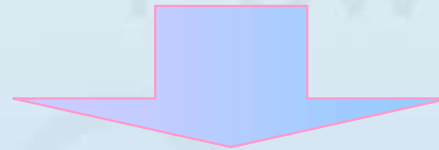
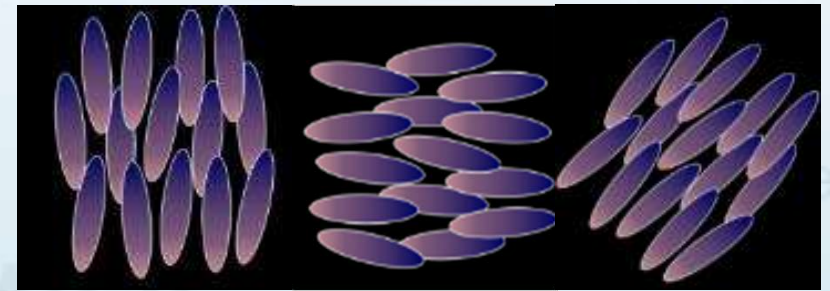
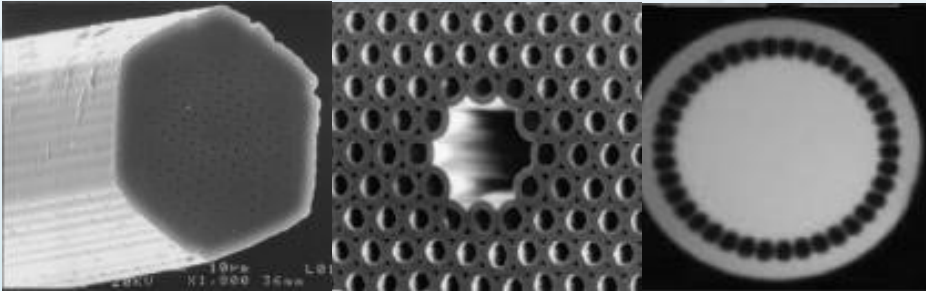
MOF

Variety of PCF structures
(birefringence, fiber modes,
dispersion, nonlinearity, etc.)



LC

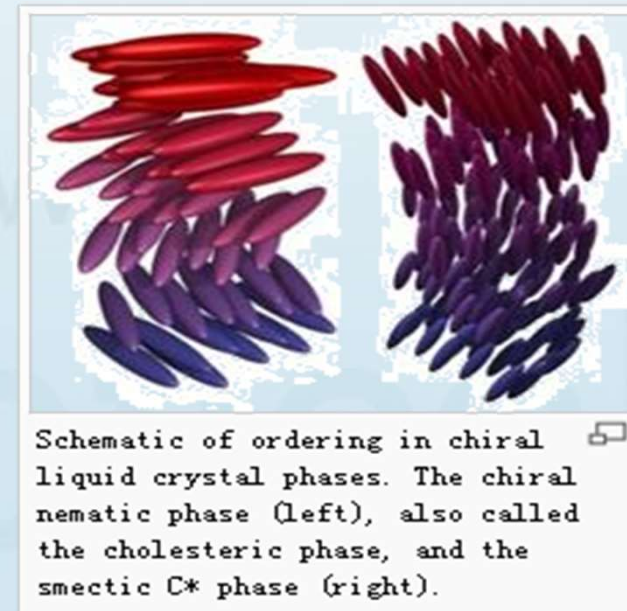
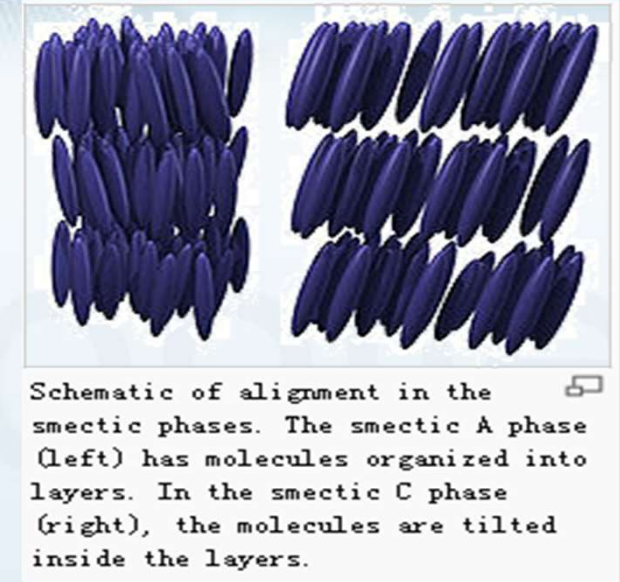
Variety of LC materials and
LC characteristics



Increase the flexibility of the waveguide properties.
More functionality of the fiber device.

2. Liquid crystal PCF: Liquid crystal phases

- Thermotropic liquid crystals
 - Nematic phases
 - Smectic phases
 - Chiral (Cholesteric) phases
 - Blue phases
 - Discotic phases
- Lyotropic liquid crystals
- Metallotropic liquid crystals



2. Liquid crystal PCF: external influences on liquid crystals

• Electrical field effects

For most LC: $-6 < \Delta\epsilon < 50$; $\Delta\epsilon = \epsilon_{\parallel} - \epsilon_{\perp}$

- Positive anisotropy $\epsilon_{\parallel} > \epsilon_{\perp}$

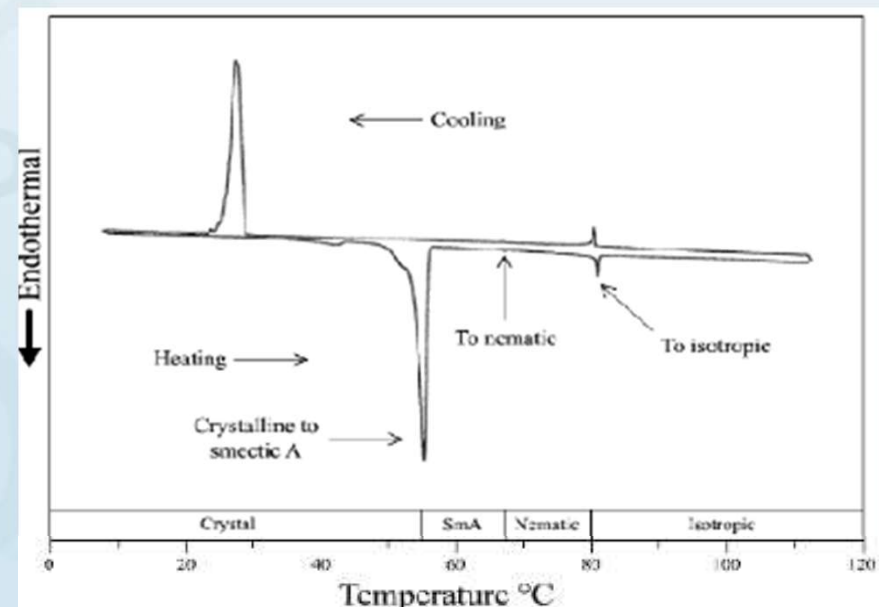
The LC would like to align parallel to the E field at high voltages.

- Negative anisotropy $\epsilon_{\parallel} < \epsilon_{\perp}$

The LC would like to align perpendicularly to the E field at high voltages.

• Thermal effects

Phase transitions can occur when temperature is varied.

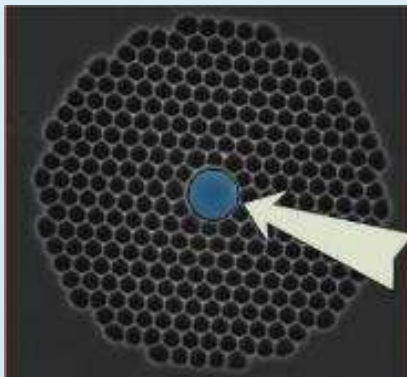


The heating process (lower) and cooling process (upper) above the temperature axis, by Prof. Charles Rosenblatt (Dept. of Physics, Case Western Reserve University)

2. Liquid crystal PCF: infiltration of LC

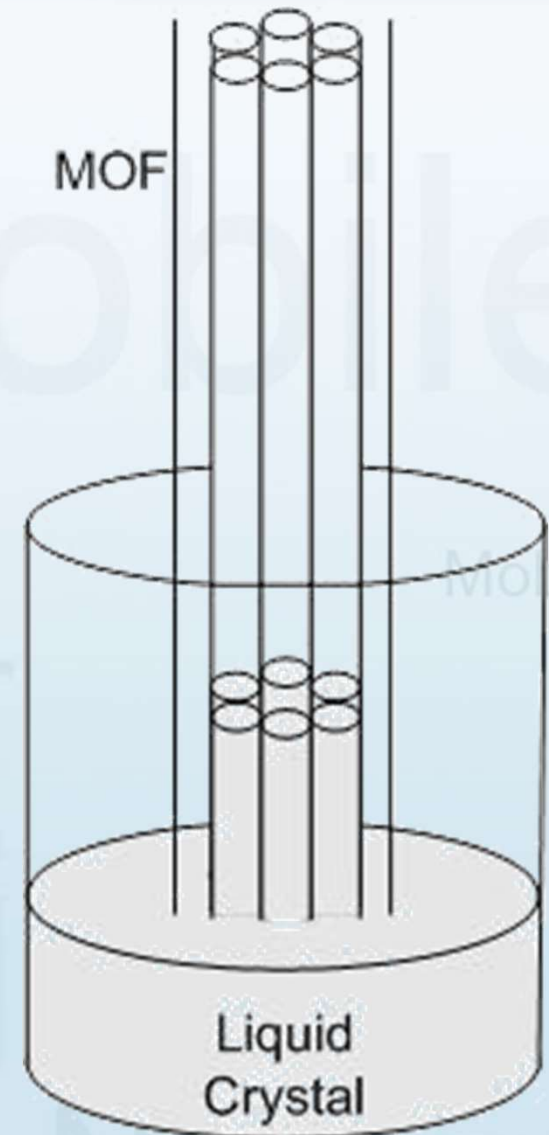
Infiltration:

- MOF is placed in container with LC
- Filling by capillary forces, vacuum or pressure
- A fusion splicer collapses smaller holes while keeping big holes open to allow selective infiltration



MOF by NKT Photonics HC-1550-02

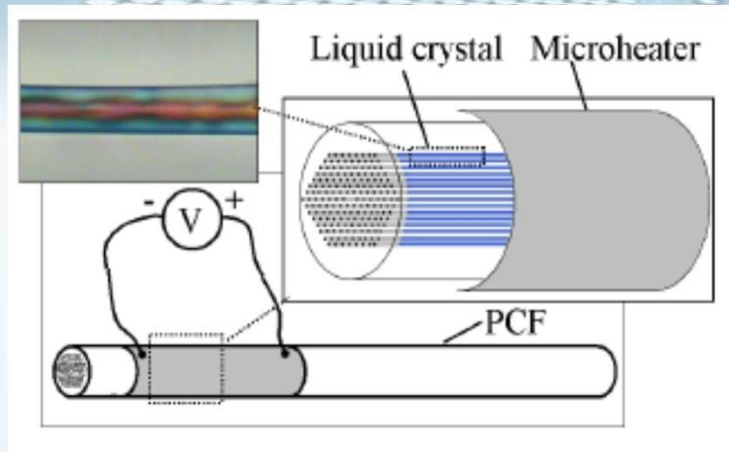
Big hole filled with liquid crystal selectively by capillary force



2. Liquid crystal PCF: applications

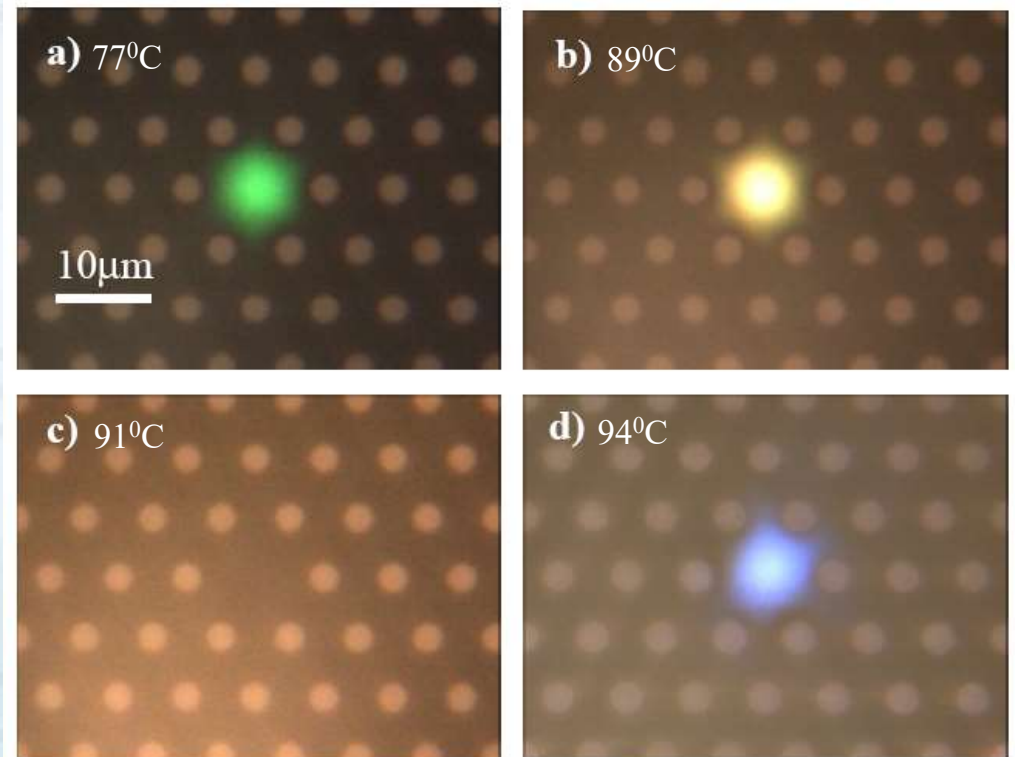
- Thermal/electrical/all-optical tuning devices, e.g. polarizer, filter, switch, modulator.
- Optical signal processing
- Nonlinear applications
- Sensing applications, e.g. temperature, bio-chemical.

2. Liquid crystal PCF: Thermo – Optic Tuning

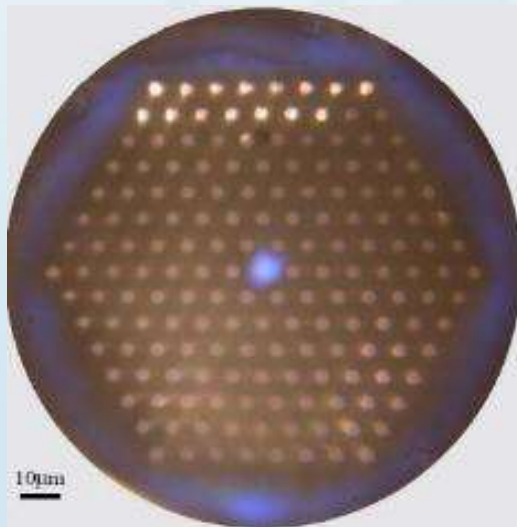


PCF filled with a short-pitch cholesteric LC

Micrographs of the guided modes in LCPCF

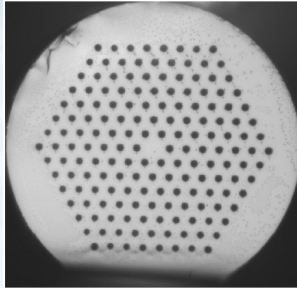


Micrographs of the guided modes in a PCF filled with a short-pitch cholesteric LC (MDA-00-1445). Well below the N* to Isotropic phase transition temperature ($T_c=94^\circ\text{C}$), the bandgap location sensitivity is approximately $1\text{nm}/^\circ\text{C}$ and $3\text{ nm}/^\circ\text{C}$ in the visible and infrared region, respectively. Closer to T_c , the transmission characteristics changes more rapidly, and the color of the guided modes changes from a) green@ $T=77^\circ\text{C}$ to b) yellow@ $T=89^\circ\text{C}$ and then into an c) off state@ $T=91^\circ\text{C}$ for thereafter to change to d) blue@ $T=94^\circ\text{C}$

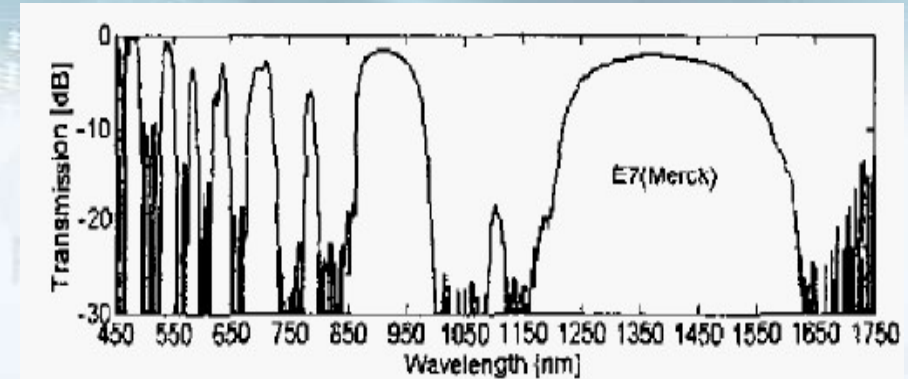


Micrograph of the end facet of a liquid-crystal-filled photonic crystal fibre. The periodic structure, consisting of silica and liquid crystal, gives rise to a cladding with photonic bandgaps, and the fibre supports only guided modes at certain wavelength bands. In this case, a blue mode is supported in the visible region of the spectrum.

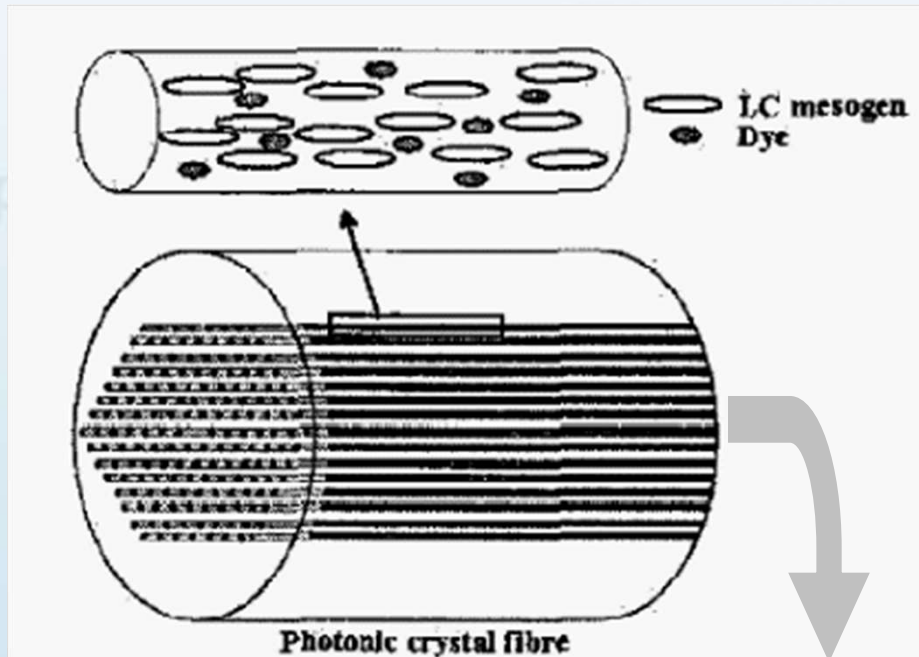
2. Liquid crystal PCF: Opto – Optic Tuning



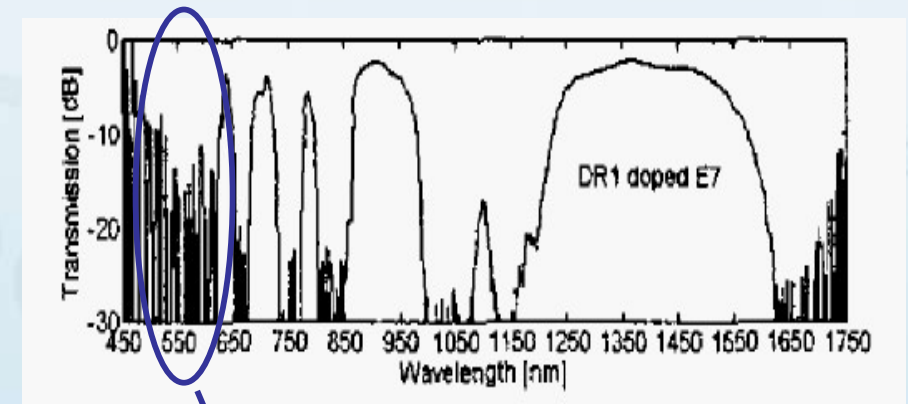
PCF
all silica
Large-Mode Area
endlessly single mode



Transmission spectrum of LCPBG fiber without dye doping



The air holes are infiltrated with a dye-doped LC (E7 merck)



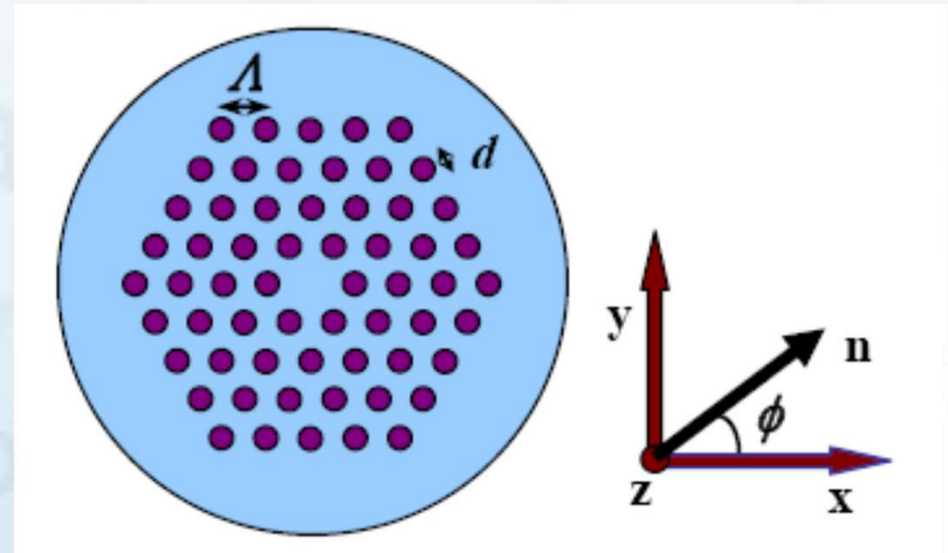
Transmission spectrum of LCPBG fiber with dye doping

Absorption around 500nm

2. Liquid crystal PCF

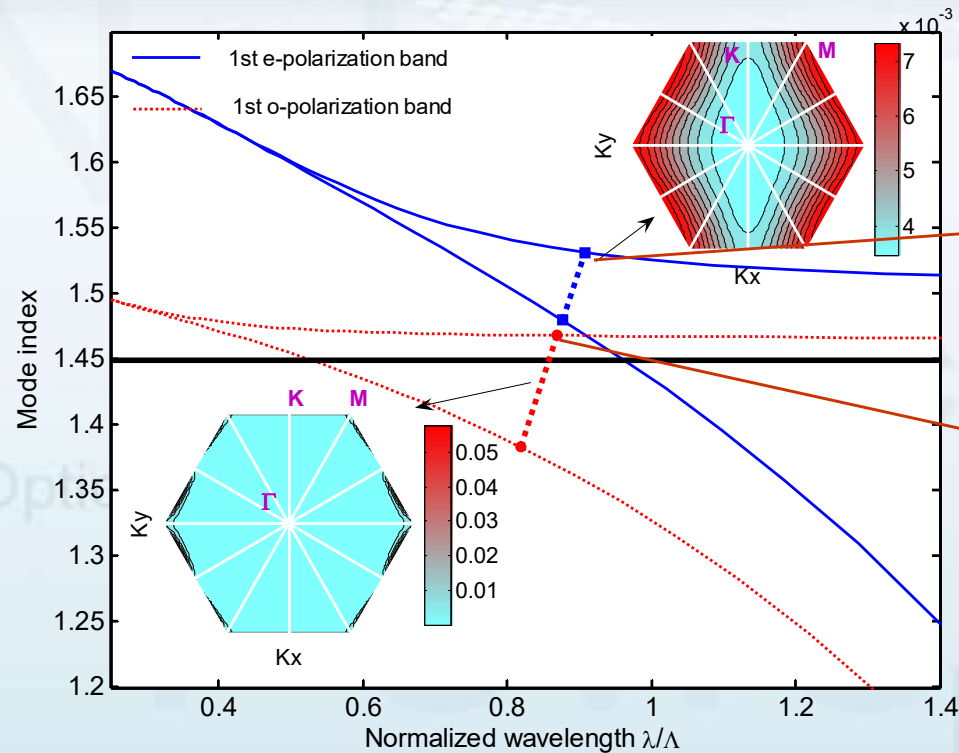
Propagation and polarization properties of photonic crystal fiber (PCF) can be manipulated by filling liquid crystal (LC). Although some paper have mentioned the guiding properties of the LCPCFs, the actual mechanism of the bandgap formation and polarized mode guiding have not been fully investigated.

- We proposed the polarization dependent bandgap formation in LCPCF.
- The polarization mode guiding of LCPCF were investigated.
- The e- or o-polarization bandgaps are responsible for mode field confinement and guiding of the e- or o-polarized modes.
- The single-mode single-polarization or ultrahigh birefringence guiding could be easily achieved in LCPCF.

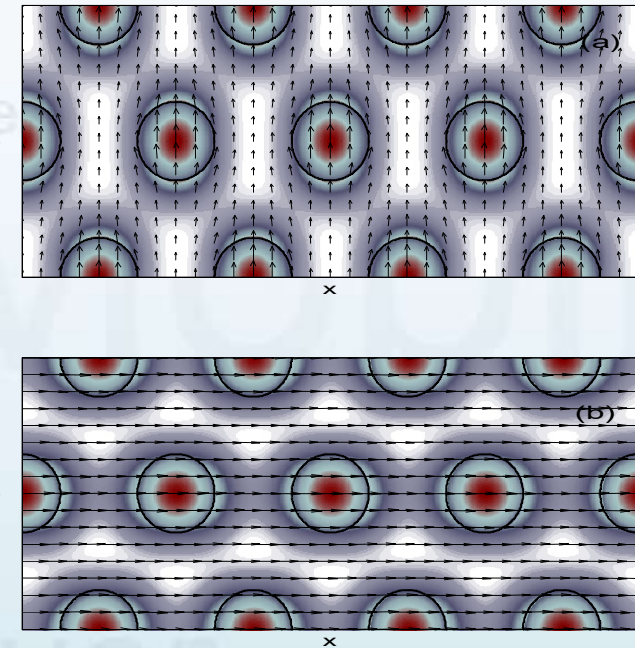


LCPCF: the purple region is infiltrated with liquid crystals. The right indicates the director of a nematic liquid crystal.

2. Liquid crystal PCF: polarization dependent bandgap formation

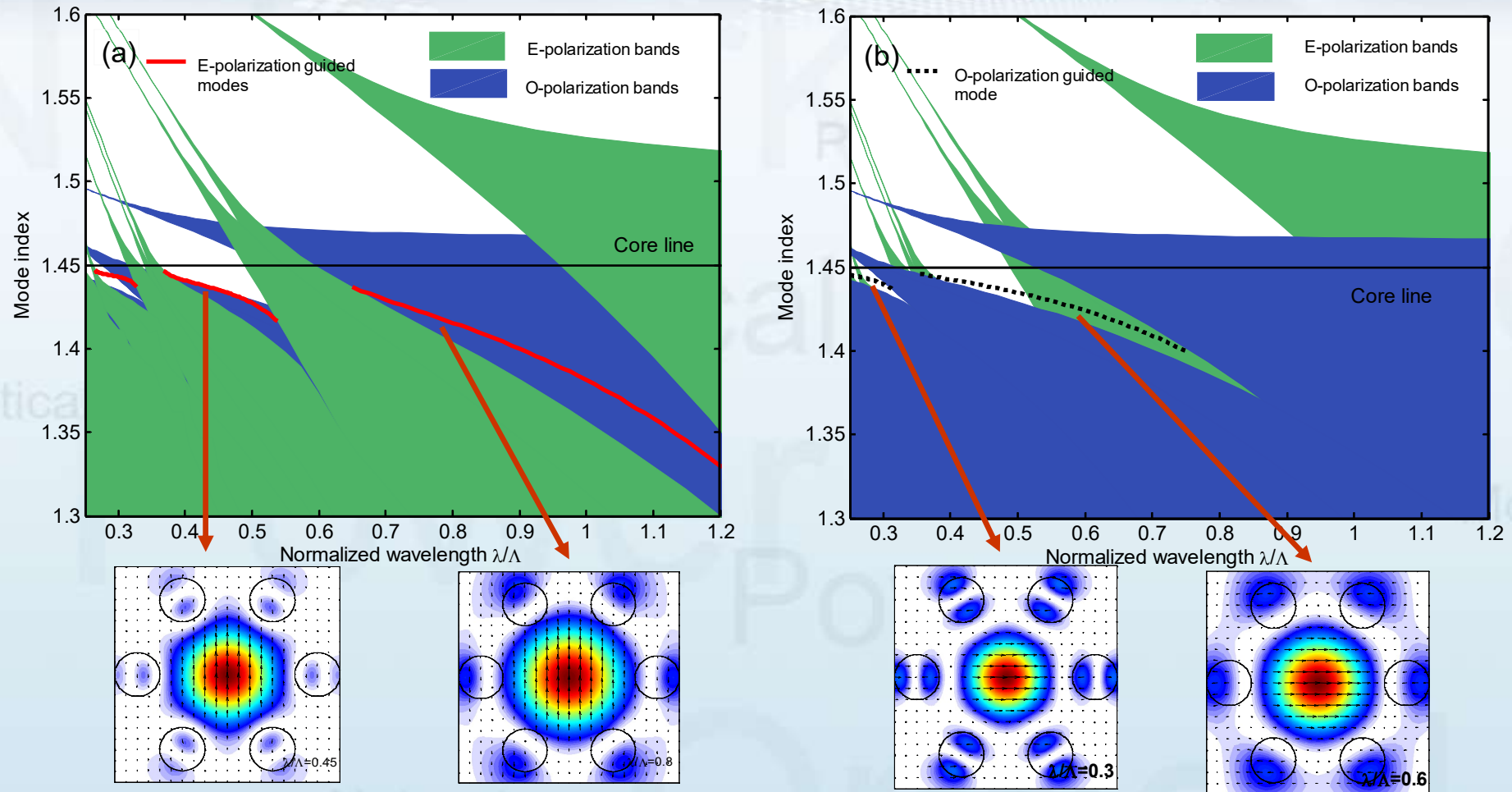


The first e and o-polarization photonic bands of LC photonic crystal cladding. The insets are the polarization ratios of e and o-polarization photonic states at normalized propagation constant $\beta\Lambda=10.6$ within the first Brillouin zone.



Electric field distributions of (a) e-polarization photonic state, and (b) o-polarization photonic state at the Γ point.

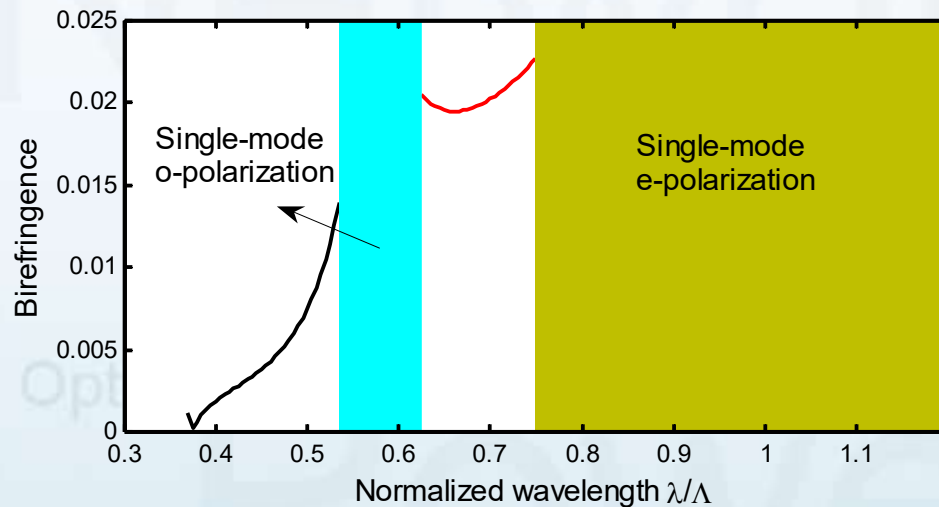
2. Liquid crystal PCF: polarization dependent guiding



Polarization dependent bandgap maps and mode dispersion curves (mode profiles) within bandgaps. (a) E-polarization guided modes (solid lines) (b) O-polarization guided modes (dot lines)

2. Liquid crystal PCF:

single-mode single-polarization or high birefringence



Birefringence (solid line) and single-mode single polarization windows of LCPCF. The LCPCF shows an ultrahigh birefringence (exceeding 0.02) around normalized wavelength $\lambda/\Lambda=0.7$.

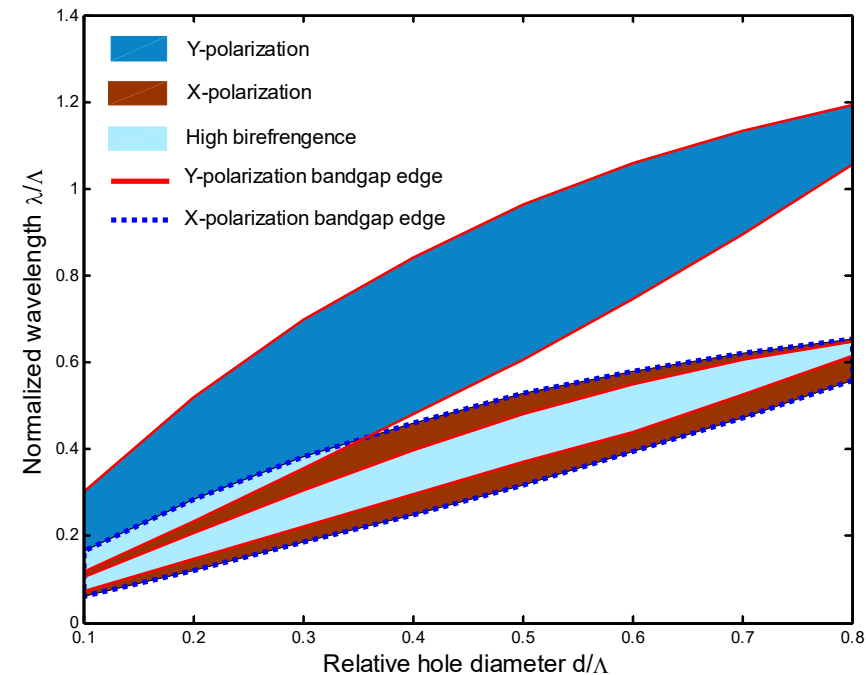
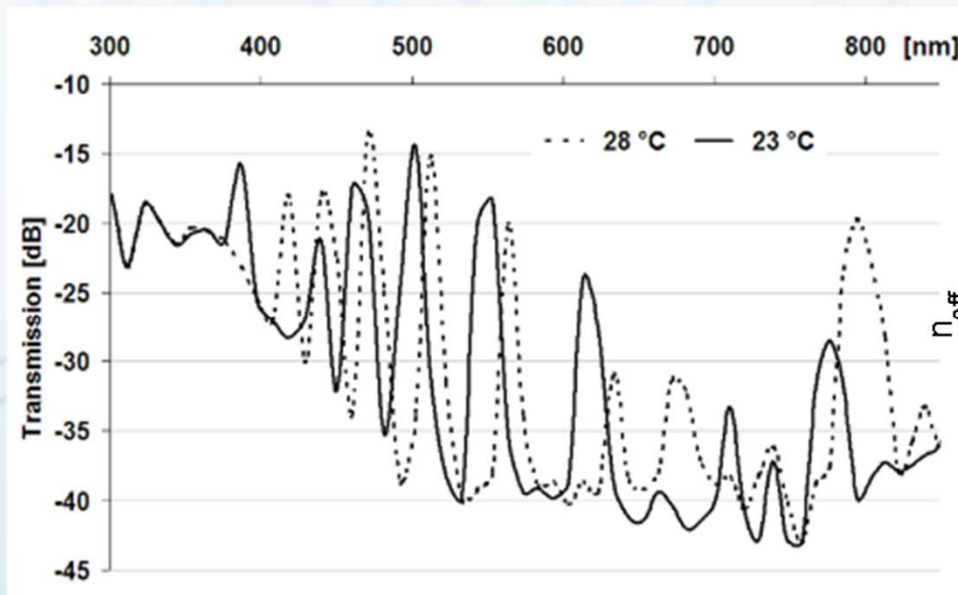


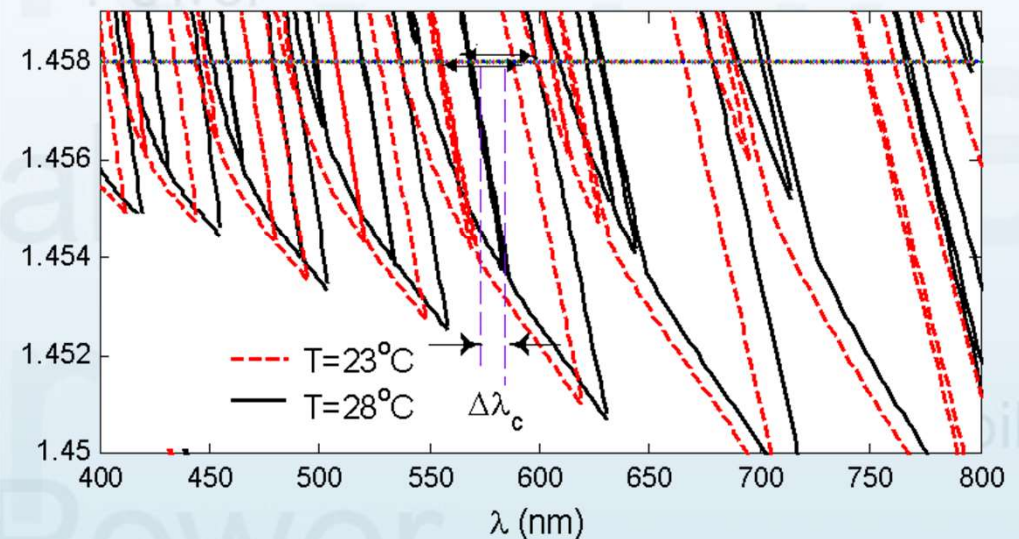
Diagram illustrating states with single-mode single-polarization or high birefringence operation of LCPCF.

2. Liquid crystal PCF:

Thermally tunable transmission bands and PBG



Experiment

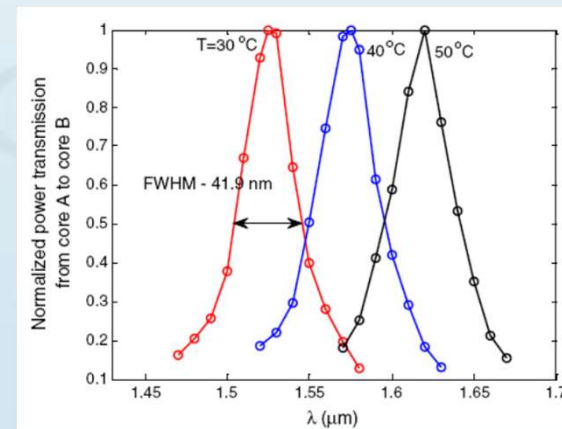
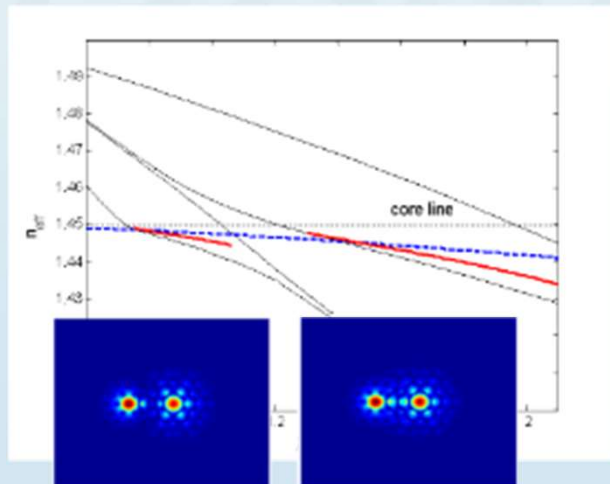
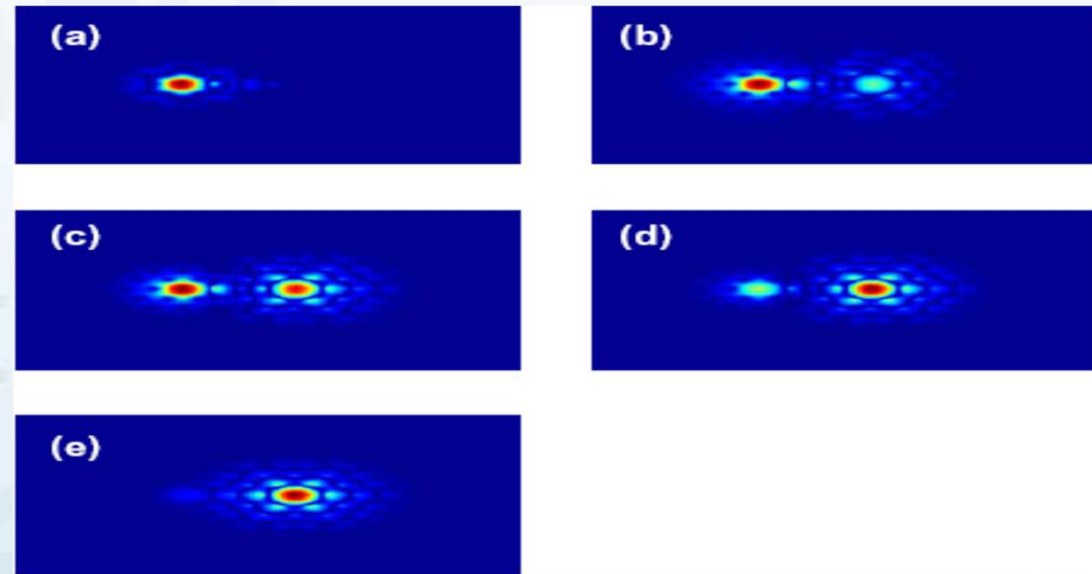
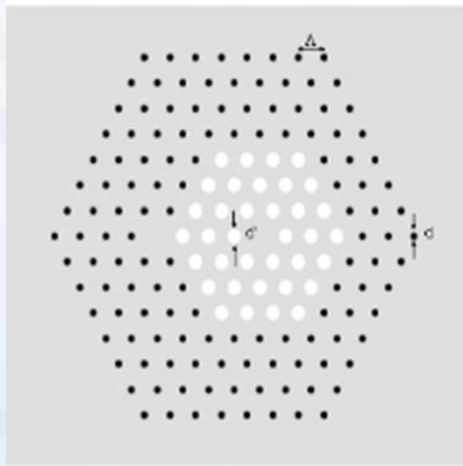


Simulation

J. J. Hu *et al.*, Optics Communications, **281**, 4339-4342, 2008
 T. R. Woliński, *et al.*, Measurement Science and Technology, **17**, 985-991, 2006.

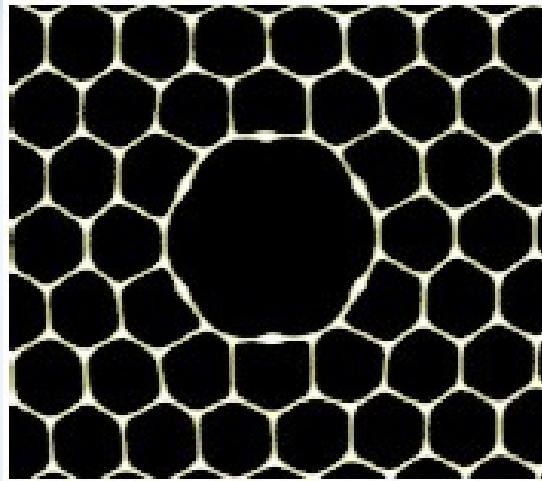
2. Liquid crystal PCF:

Design and Analysis of a thermally tunable PCF coupler with NLC infiltration

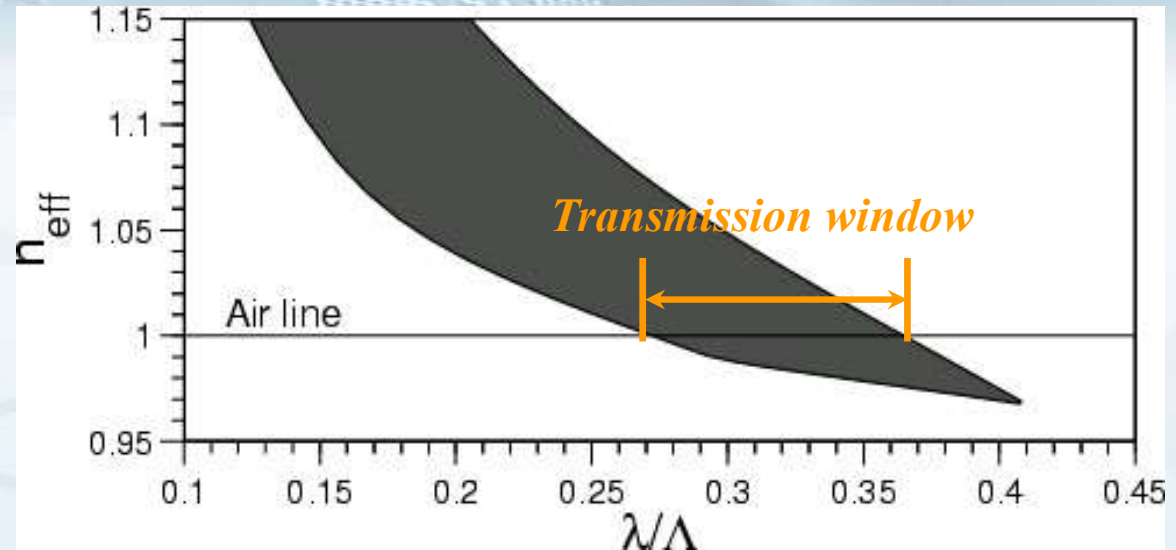


D. J. J. Hu *et al.*, Optics Communications, **282**, 2342-2347, 2009

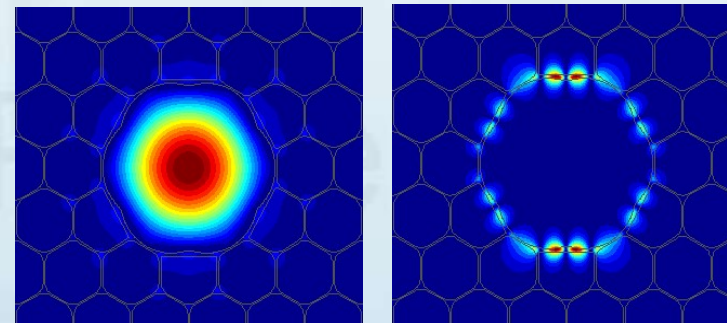
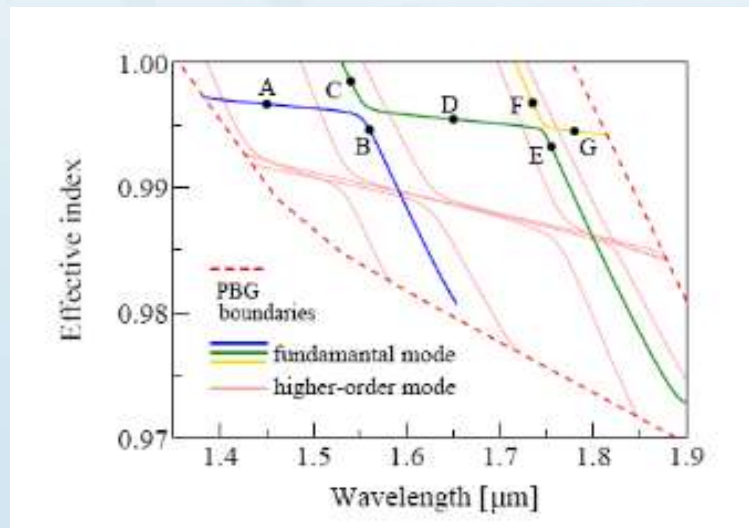
3. Improved Triangular-lattice PCF



Corning Inc., 2003



Transmission window: ~440nm



A

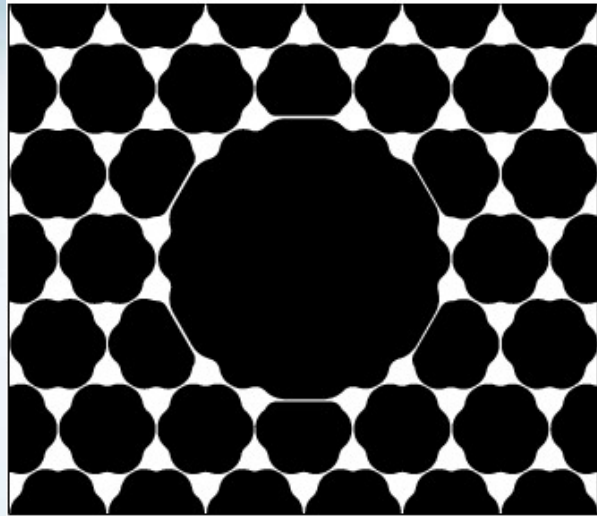
B

Shortcomings: the surface modes have a strong impact on dispersion and leakage properties of the fundamental mode.

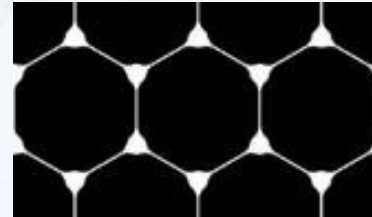
Nature **424**, p.657 (2003).

Opt. Express **12**, p. 394 (2004). 33

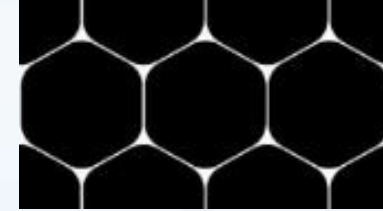
3. Improved Triangular-lattice PCF



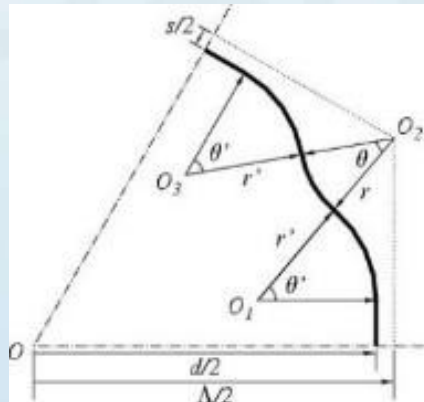
~570nm



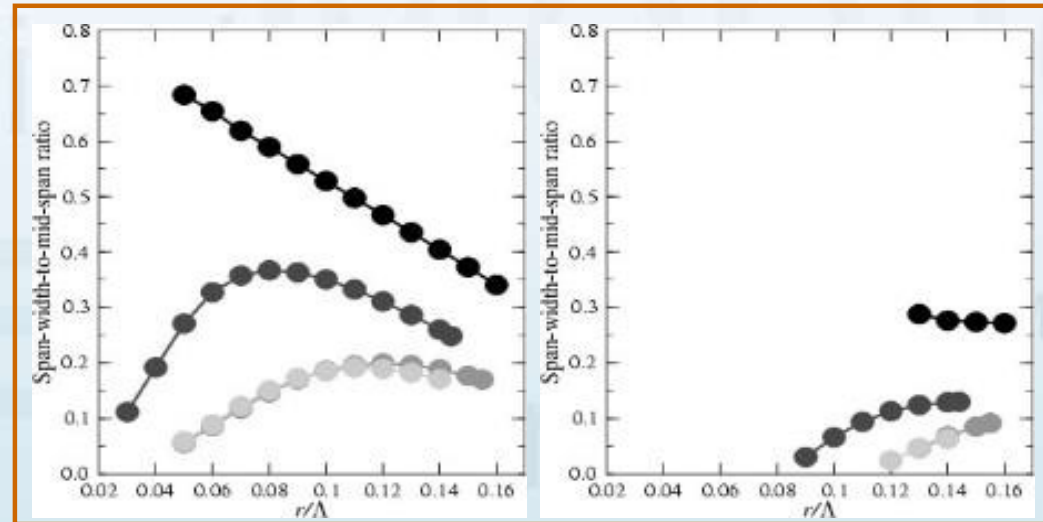
~440nm



$\Lambda = 2.6 \mu\text{m}$,
 $s = 0.02\Lambda$,
 $r = 0.14\Lambda$,
 $\theta = 40^\circ$.

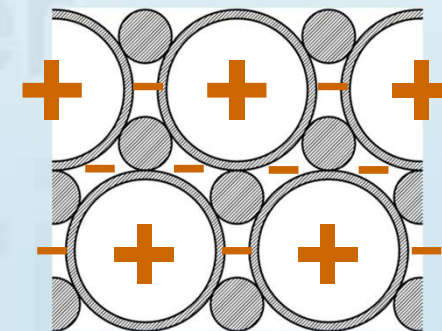
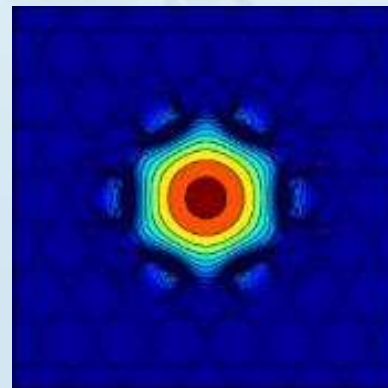
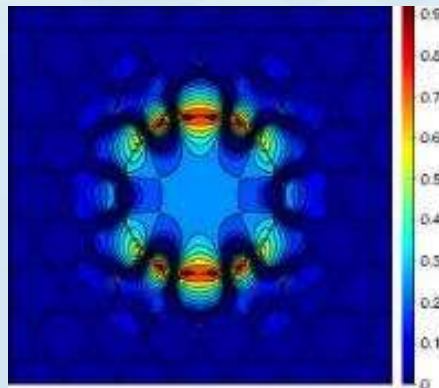
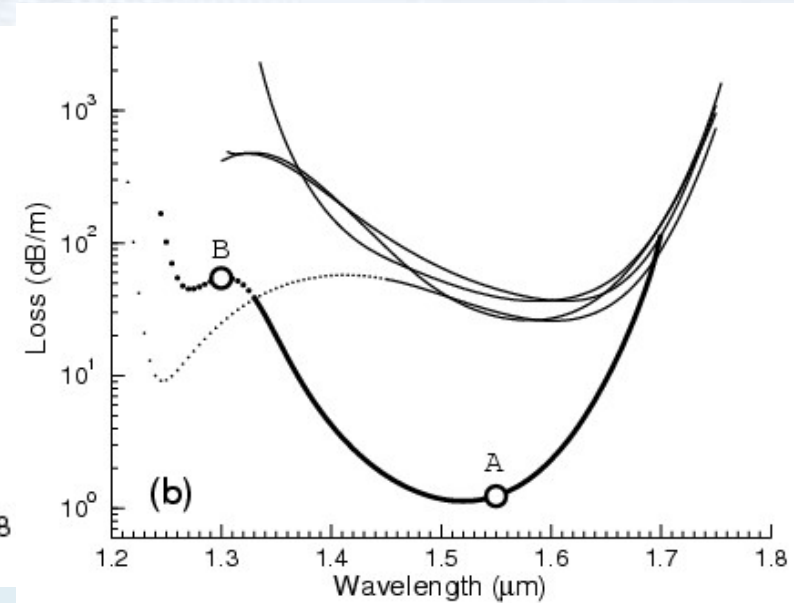
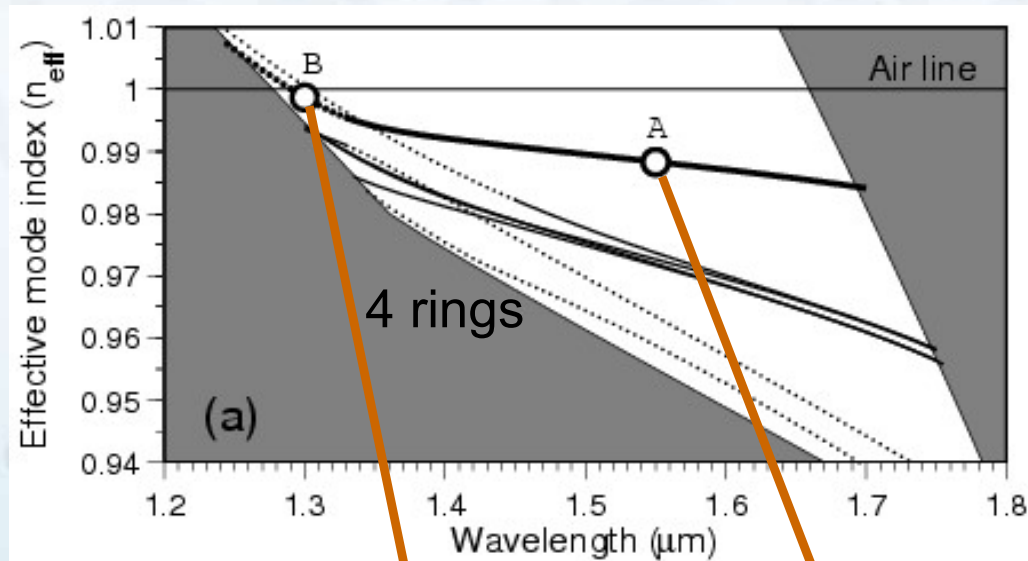


key parameters: r, s, θ



This type of cladding would perform better in terms of the PBG-guiding wavelength range, radiation loss owing to finite cladding size, and the ability to avoid surface mode problems.

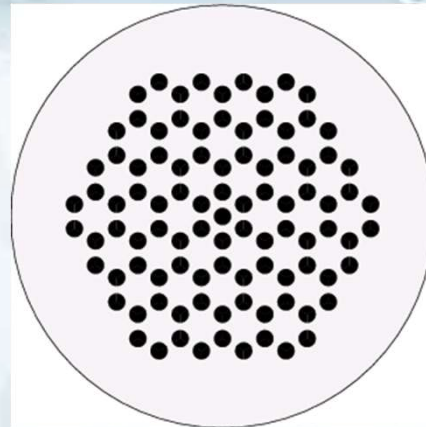
3. Improved Triangular-lattice PCF



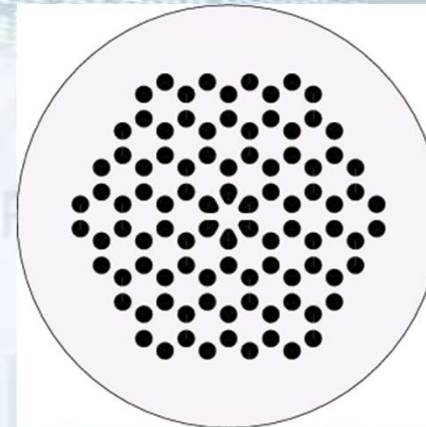
Preform stacking & Pressurization

4. Air-guiding Honeycomb Bandgap Fiber

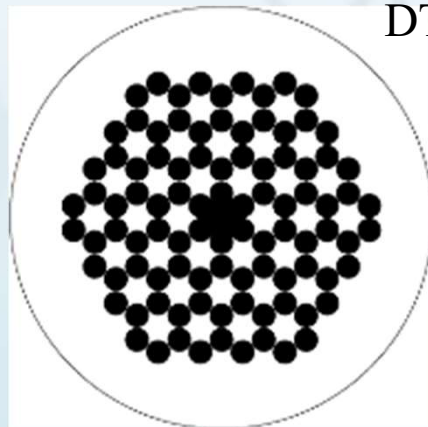
Design evolution



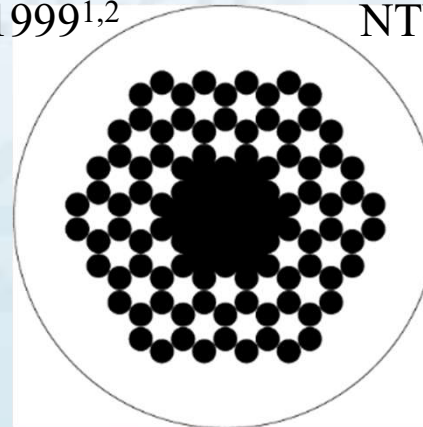
DTU, 1999^{1,2}



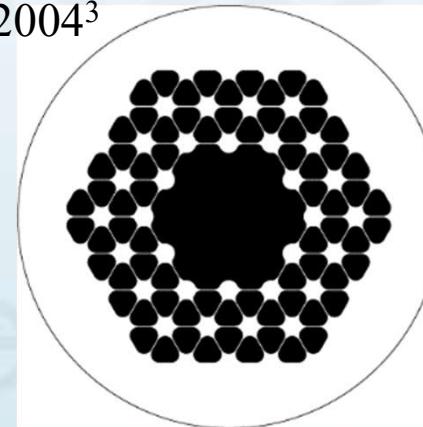
NTU, 2004³



NTU, 2005⁴



NTU, 2005⁴

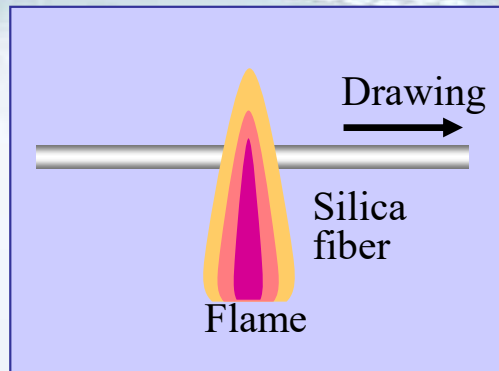


NTU, 2005⁵

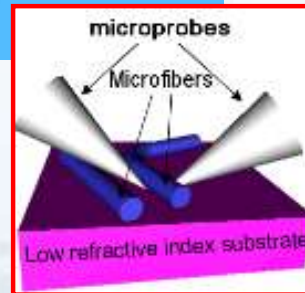
1. J. Broeng, et al., J. Opt. A **1**, p.477 (1999).
2. J. C. Knight, J. Broeng, et al., Science **282**, p.1476 (1998).
3. M. Yan et al., IEEE Photon. Technol. Lett. **16**, p.2051 (2004).
4. M. Yan et al., IEEE Photon. Technol. Lett. **17**, p.64 (2005).
5. M. Yan et al., Opt. Lett. **30**, p. 465 (2005).

5. Nanofiber

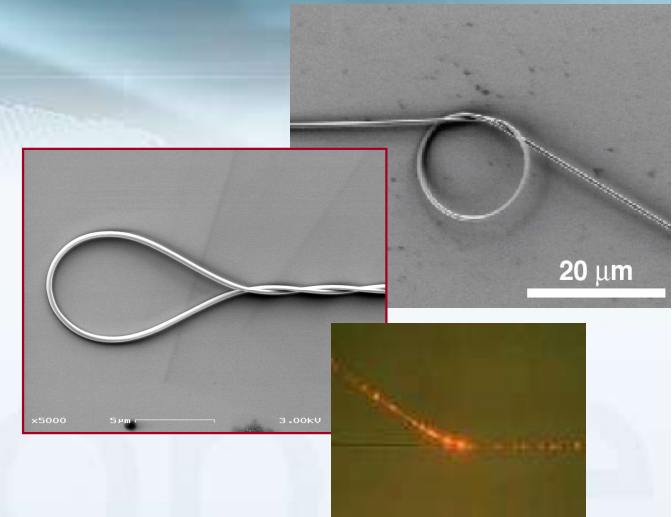
The fabrication of nanofibers



Flame-drawing or
CO₂-drawing method

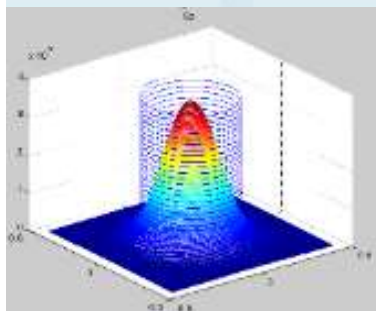


Manipulating using microprobes
and translation stages

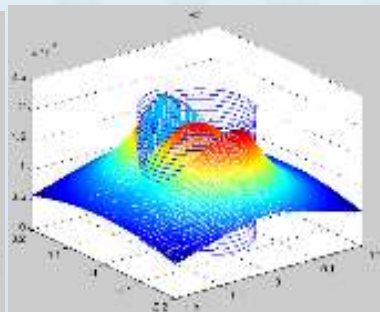


Nanofiber devices for sensing
and communications (coupler,
loop mirror, knot resonator...)

Properties of nanofibers and their devices



Strong guiding
ability ($D/\lambda > 0.7$)

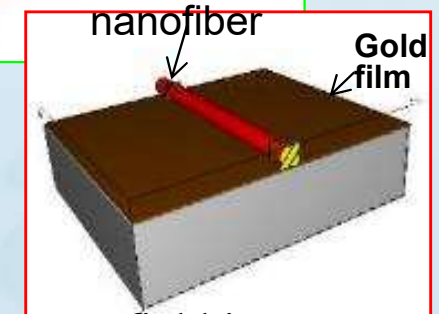
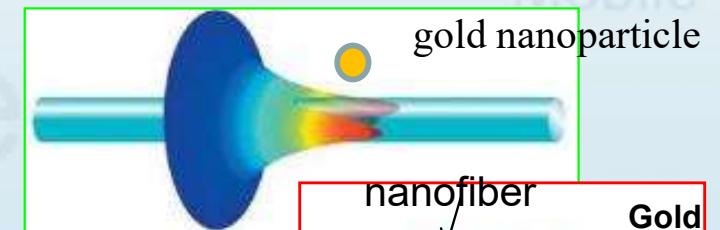


Strong evanescent
field ($D/\lambda < 0.7$)

Trade off between guiding ability
and evanescent field

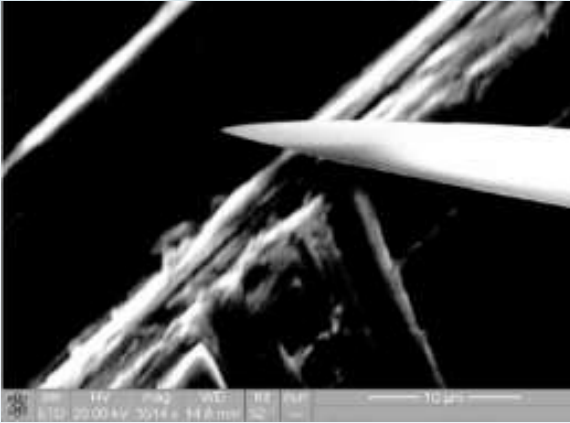


High Q nanofiber resonator.
($Q = 4.6 \times 10^5$)

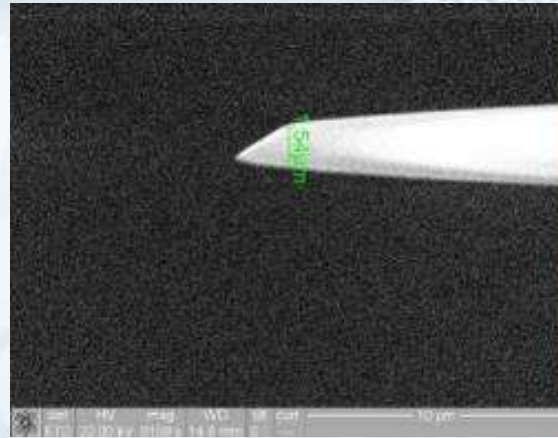


Strong evanescent field interacts
with metal surface and particles

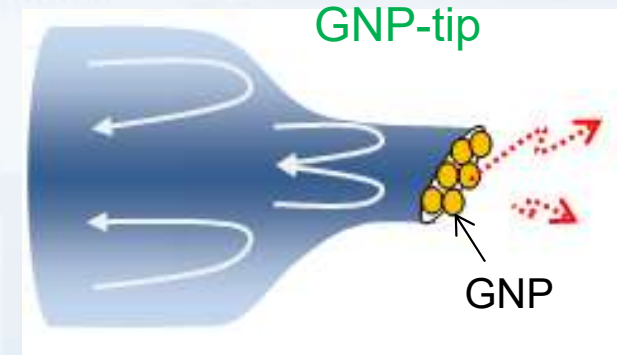
Live Cell Index Sensing Using Tilted Nanofiber Tip with Gold Nanoparticles (GNP)



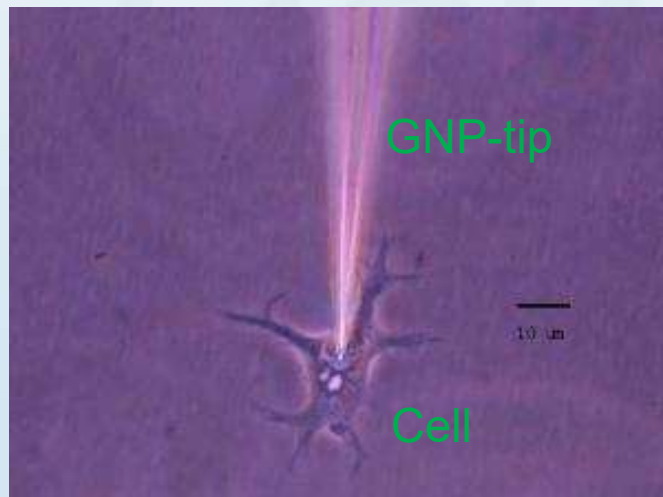
Fabrication of nanofiber tip ($<1\mu\text{m}$)



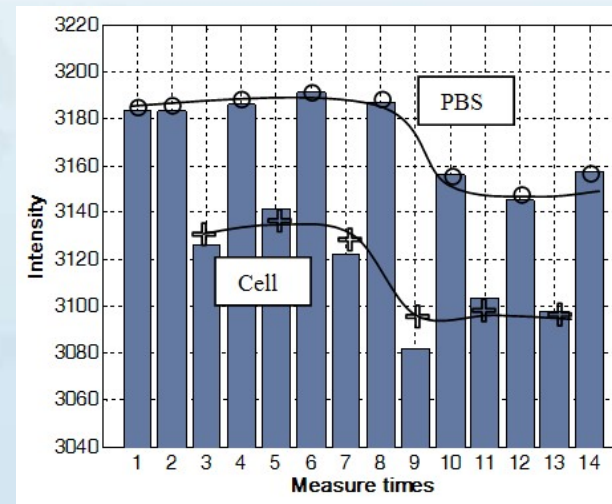
FIB cutting with an angle



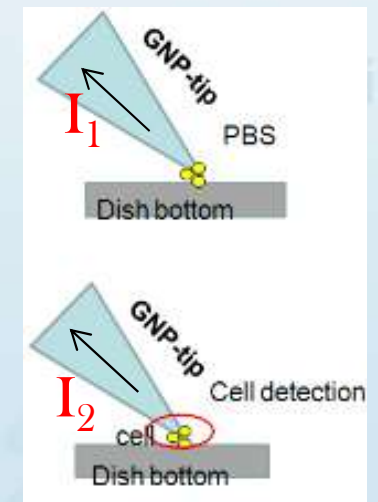
Coated with GNP at tip apex to enhance the tip reflection



Thrust into live cell for label free bio- measurement

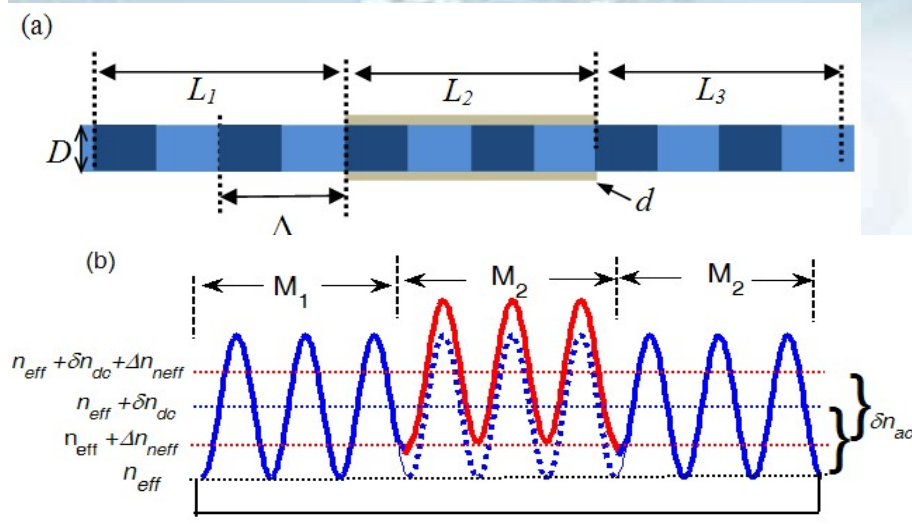


measurement of refractive index of Live cell



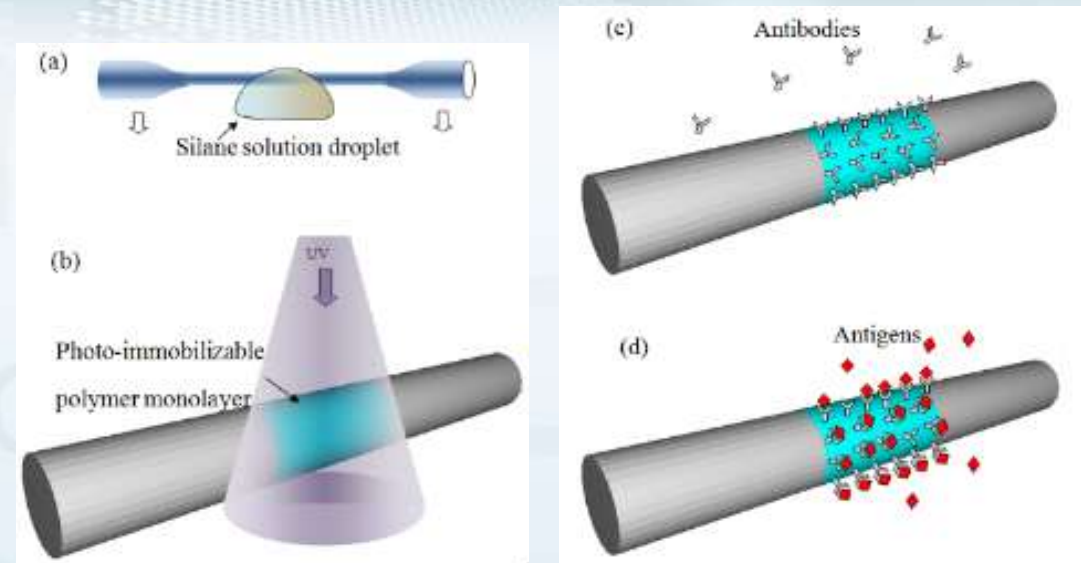
reflection difference $I_1 - I_2 \propto n_{cell} - n_{PBS}$ index difference

Localized Biosensing and Index Sensing by Introducing Effective Phase Shift in Microfiber Bragg Grating (μ FBG)

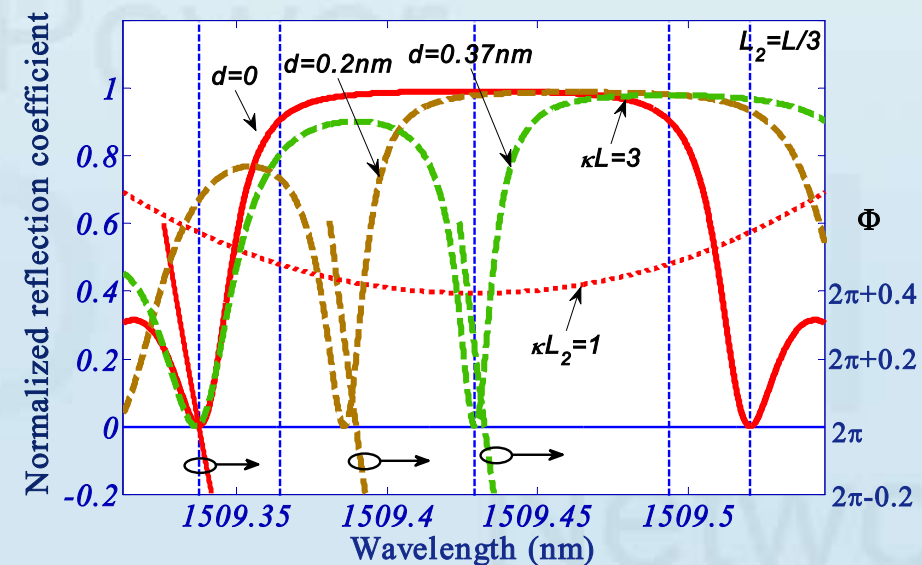


Index perturbation at the center segment, which induced effective phase shift.

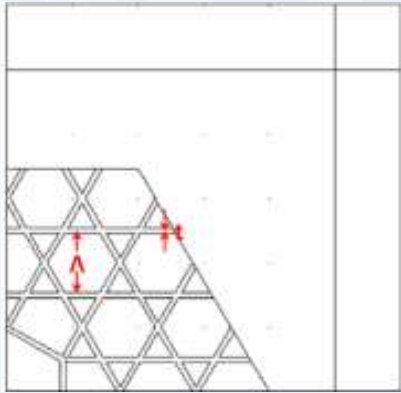
For different coating thickness, or different mass density, the dip in the reflection spectrum shift.



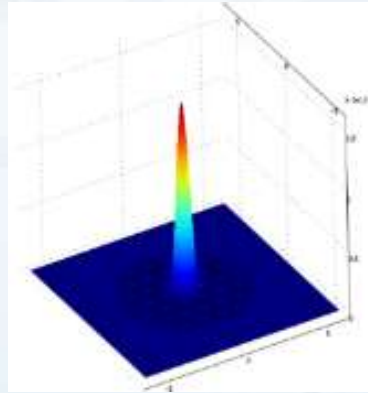
Biomolecular attachment induced phase shift for biosensing measurement



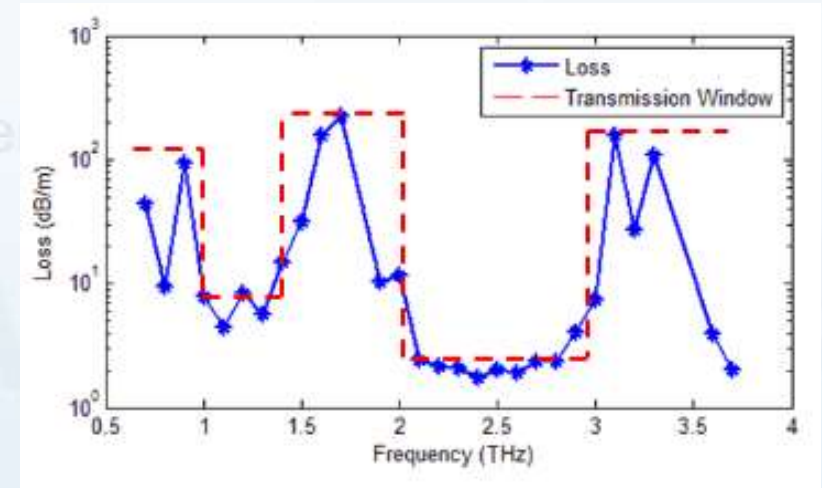
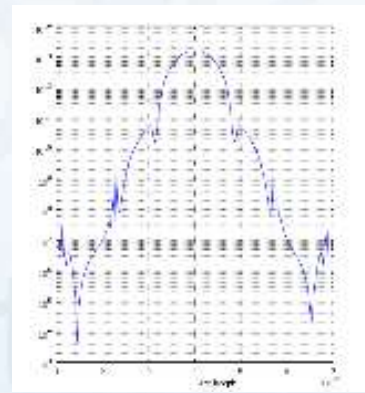
6. Terahertz kagome-lattice hollow core fibres



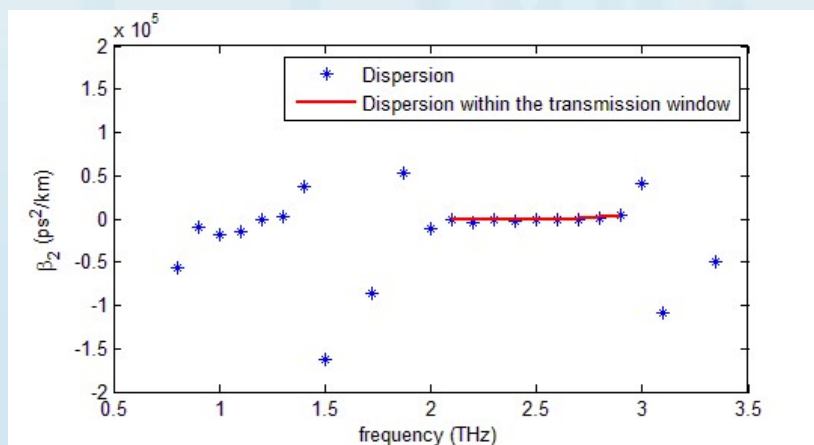
Fibre design



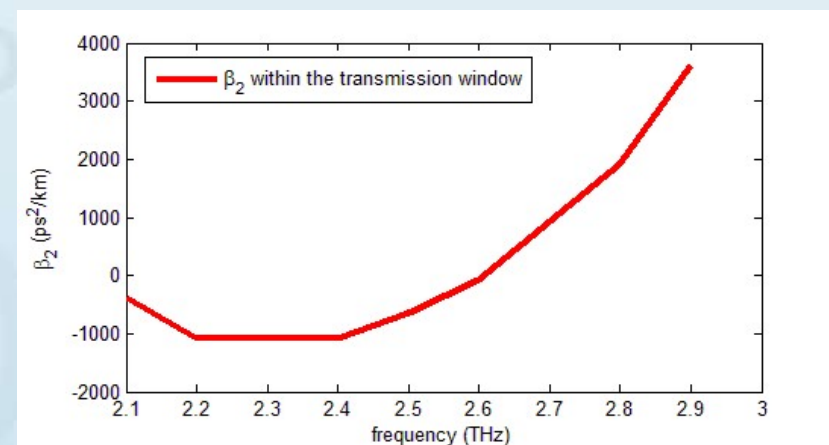
Mode profile and intensity



Loss profile

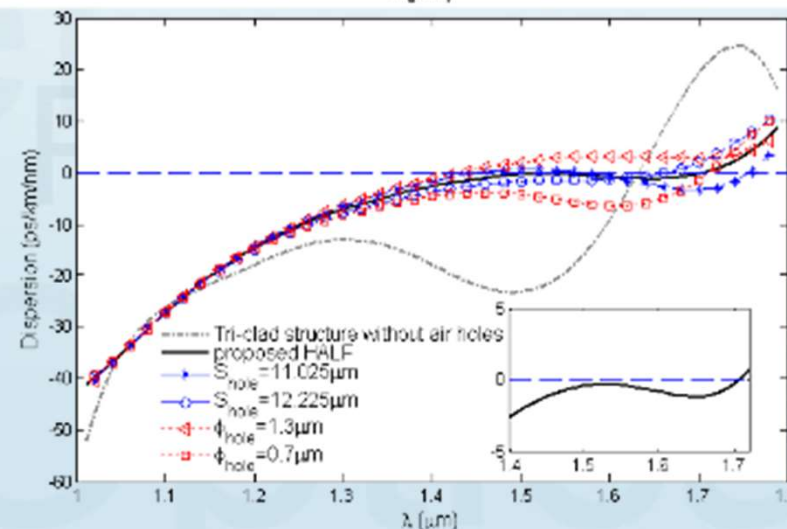
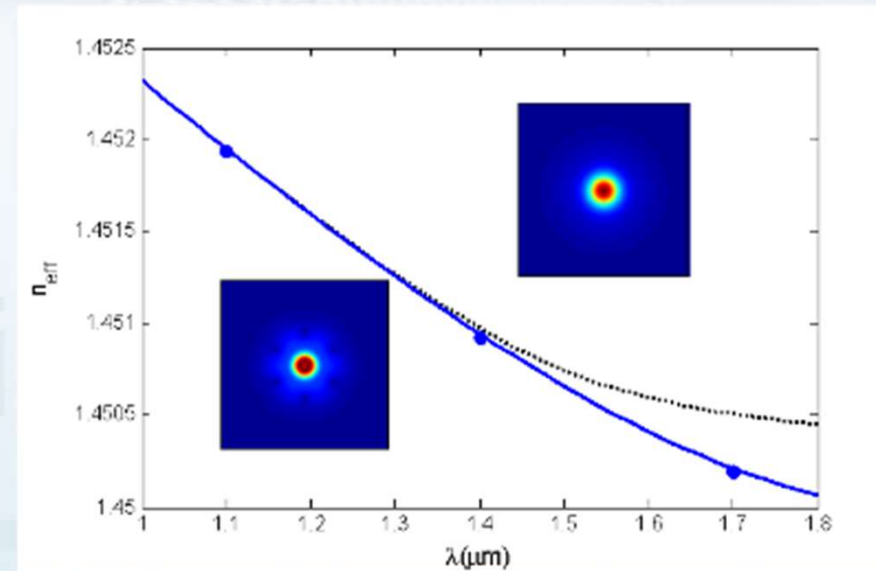
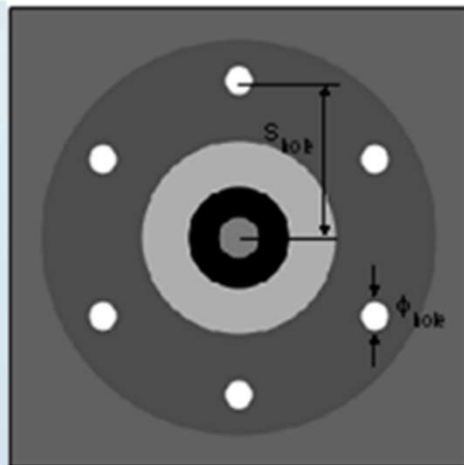
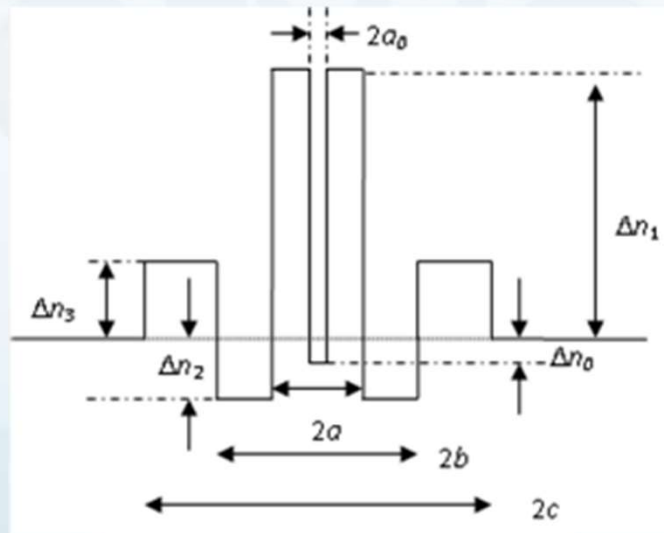


Dispersion



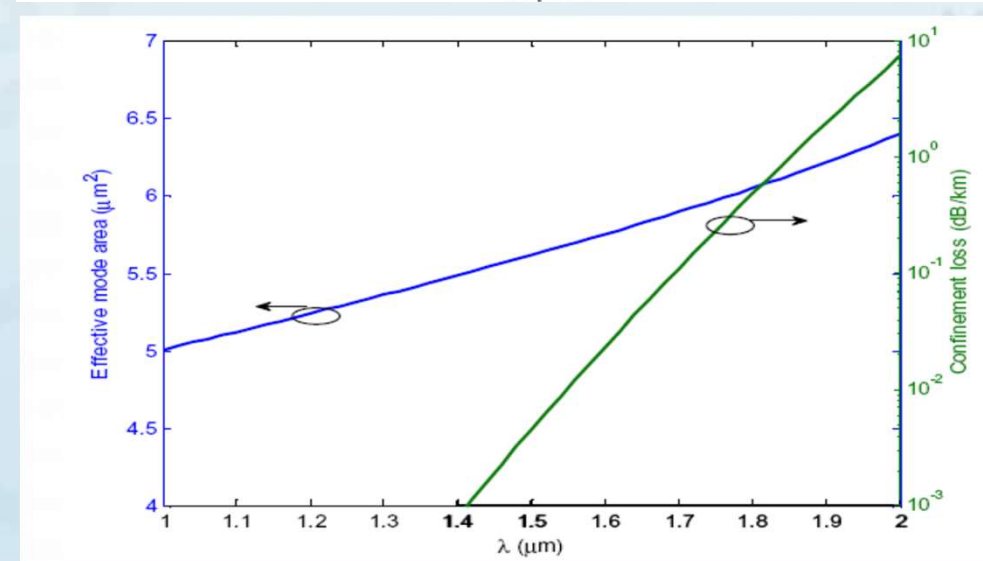
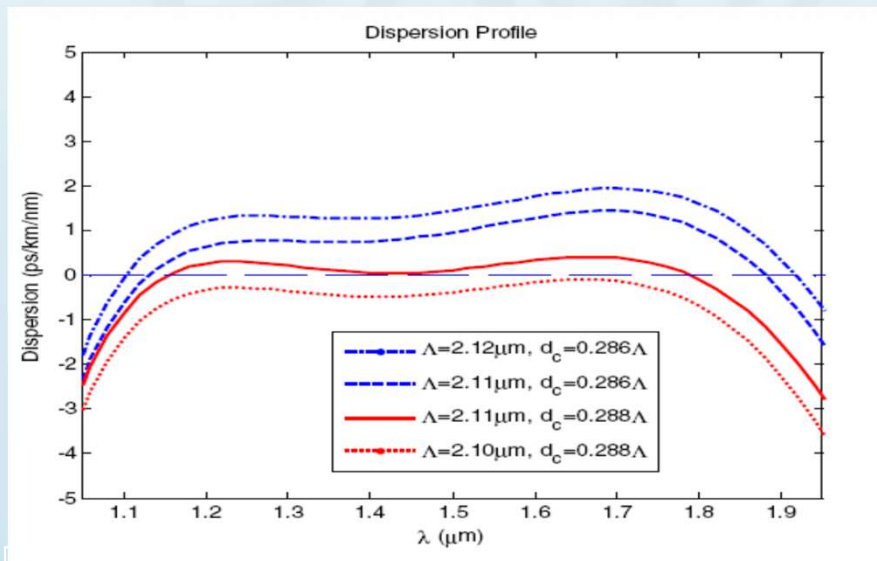
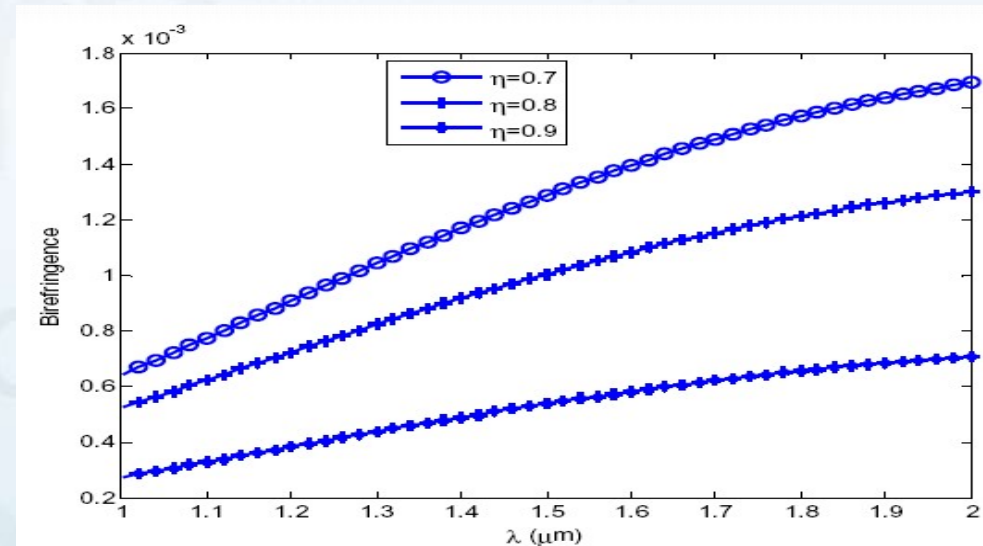
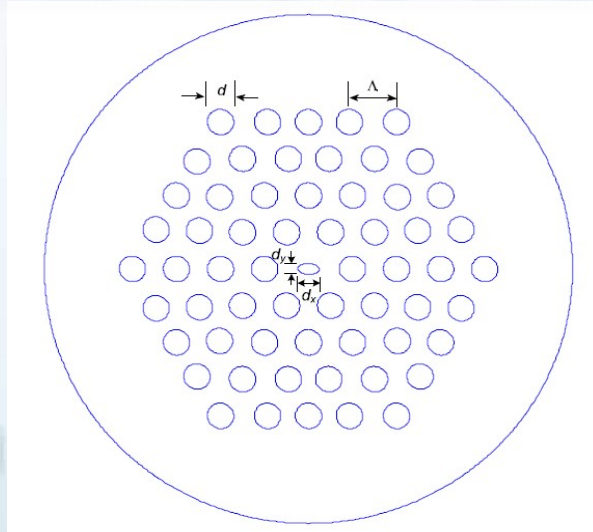
Dispersion-zoom in

9. Hole-Assisted Lightguide Fibers (HALF)

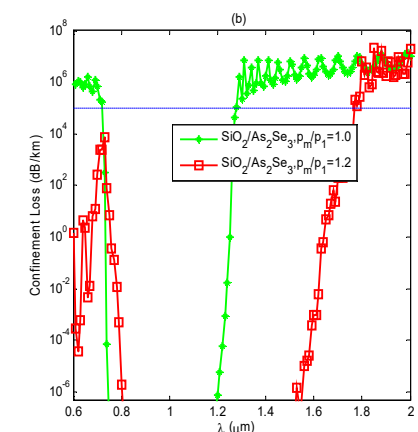
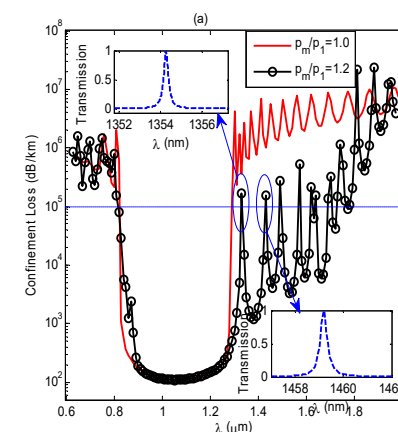
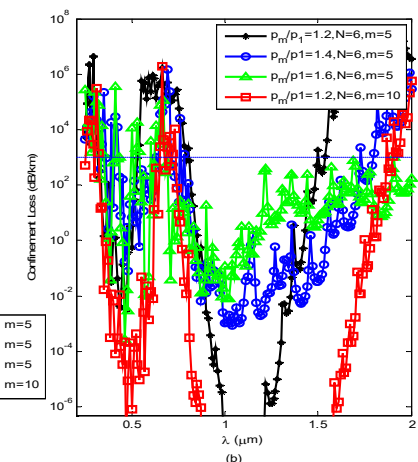
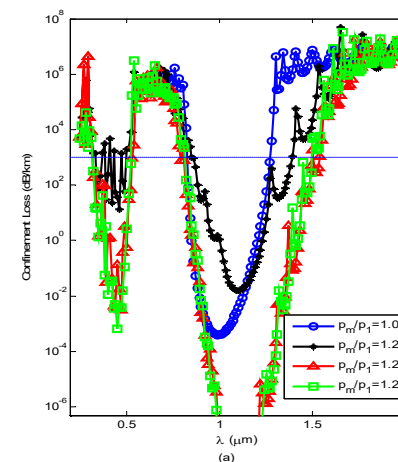
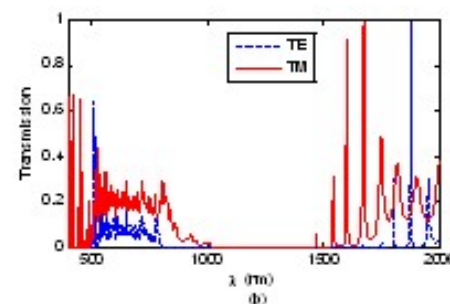
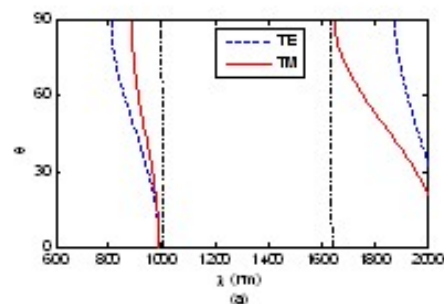
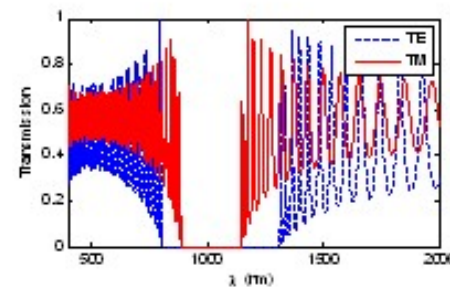
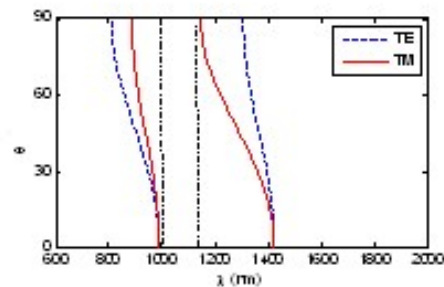
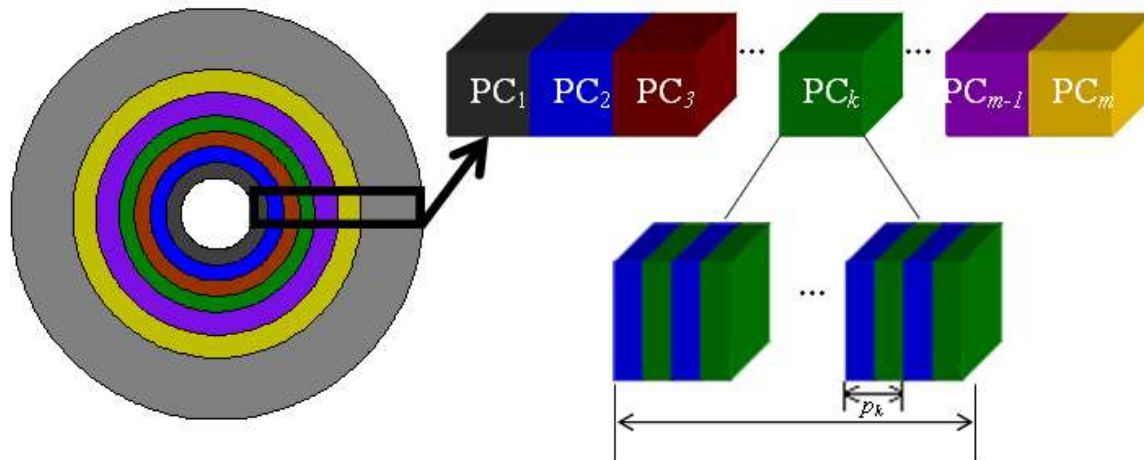


D. J. J. Hu *et al.*, Applied Optics, **47**, 5061-5064, 2008

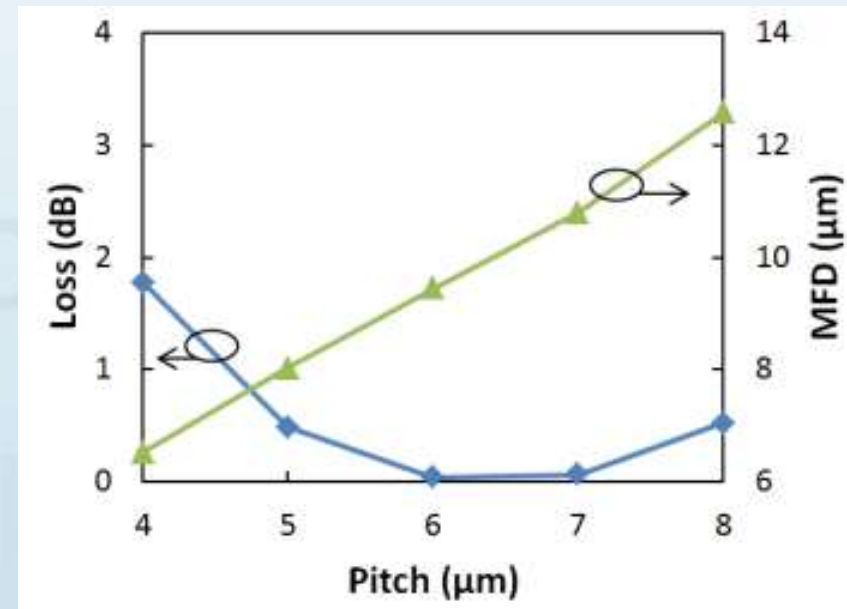
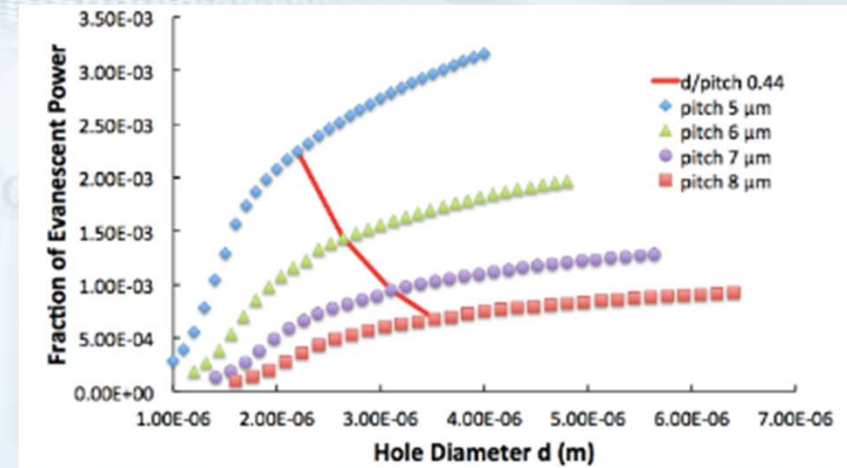
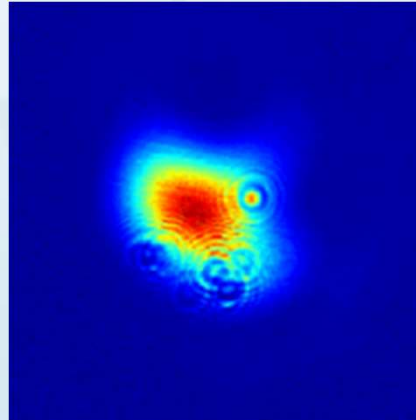
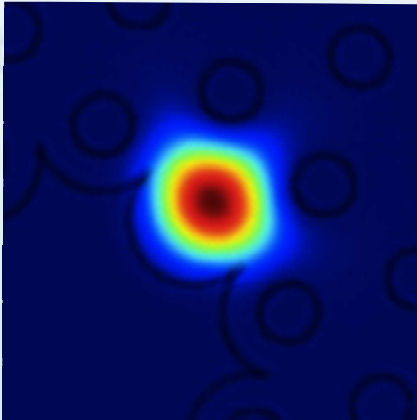
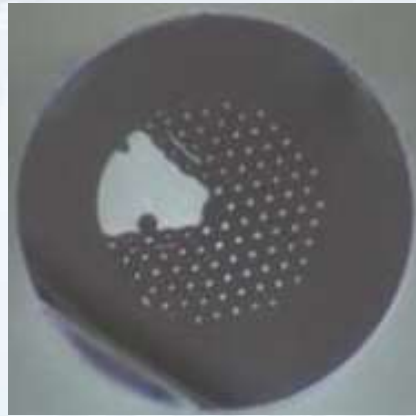
10. Dispersion-flattened PMPCF



12. Broadening of the guiding range of a Bragg fiber with geometrically distributed multilayered cladding



13. Side-channel PCF



Advantages:

- Robust after etching
- Having a channel to support the flow of liquid
- Long interaction length

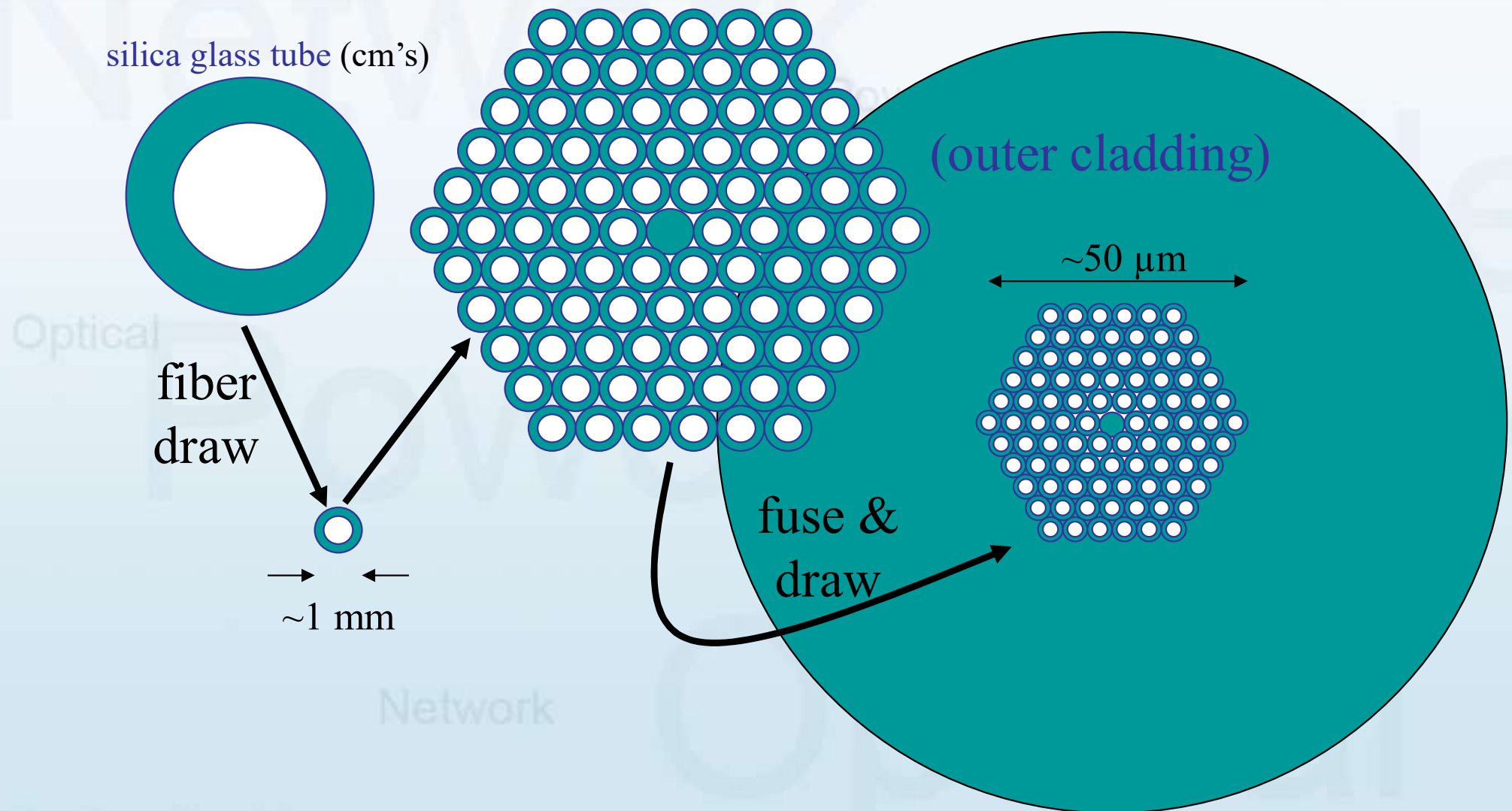
II. MOF Fabrication

1. Twin-core polymer MOF
2. Nanostructure core fiber (NCF)
3. All-solid photonic bandgap fiber (PBGF)

Fabrication facilities:

- Optical fiber drawing tower
- MCVD (PCVD) lathe (Platform for fiber preform fabrication)
- Preform drilling machine (for Polymer MOF)

Fabrication of MOF

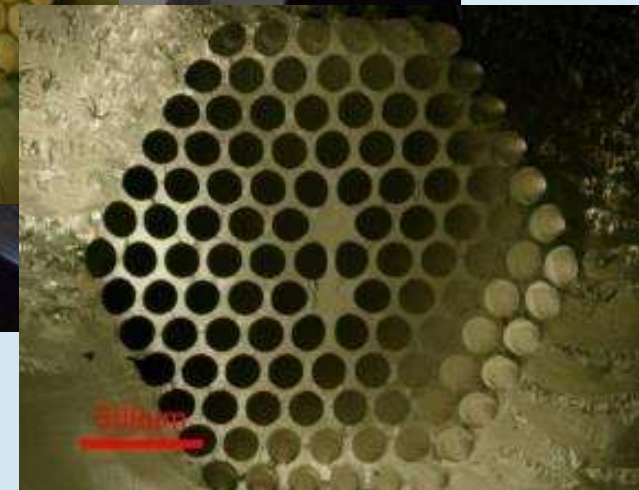
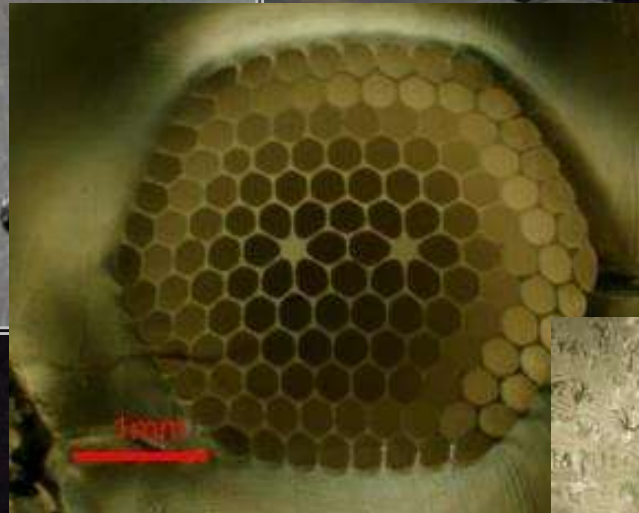
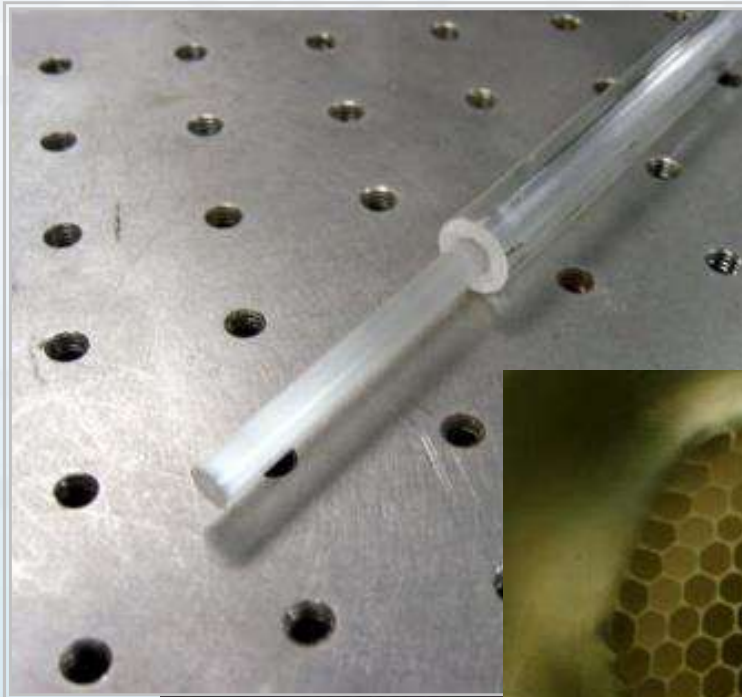


Preform Drilling Machine and Polymer Fiber Drawing Tower

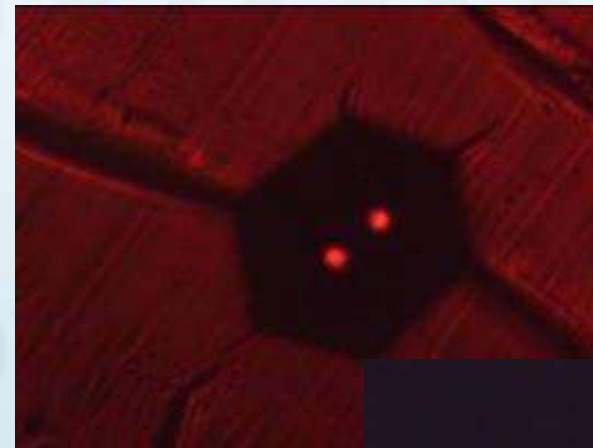
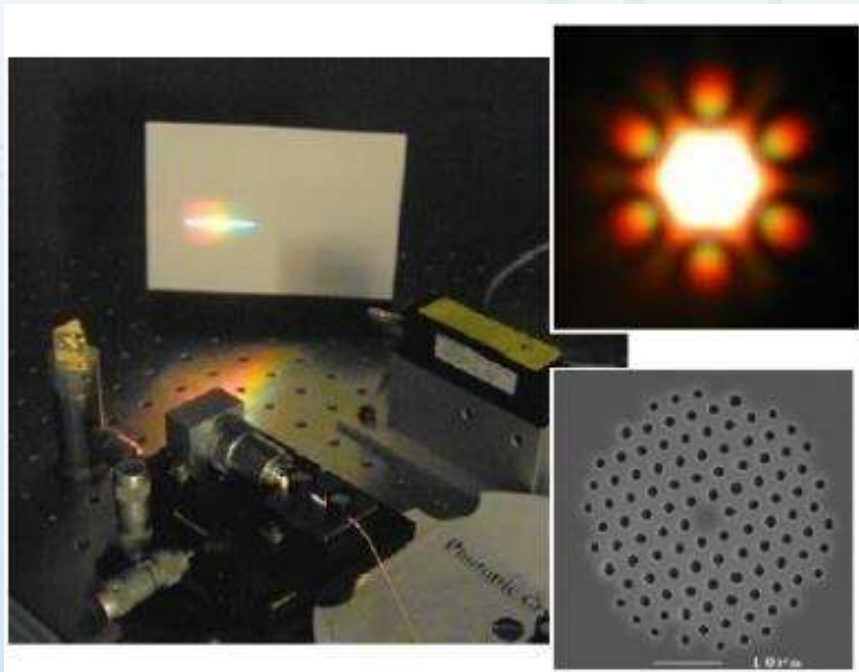
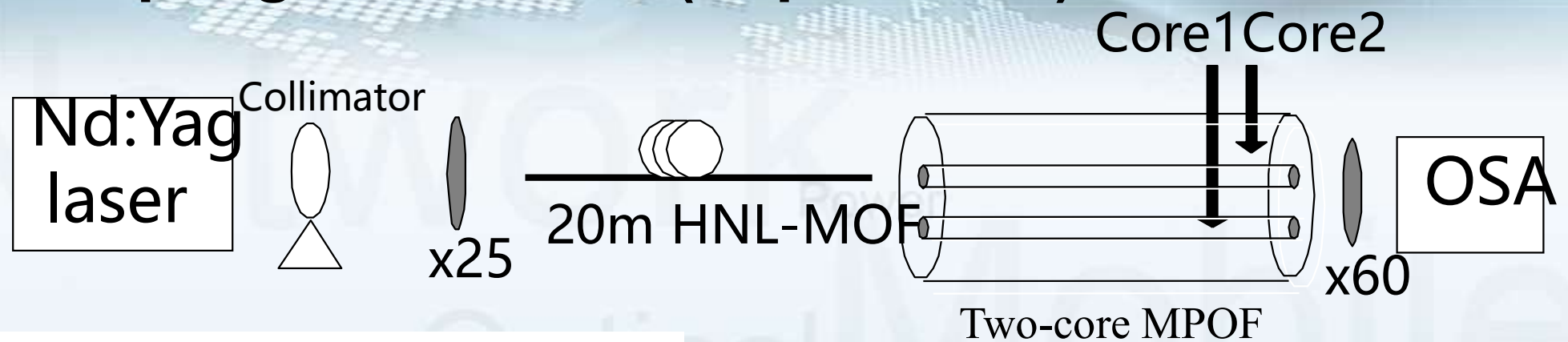


1. Fabrication of Twin-core polymer MOF three-step process

preform drilling => cane sleeving => fiber drawing



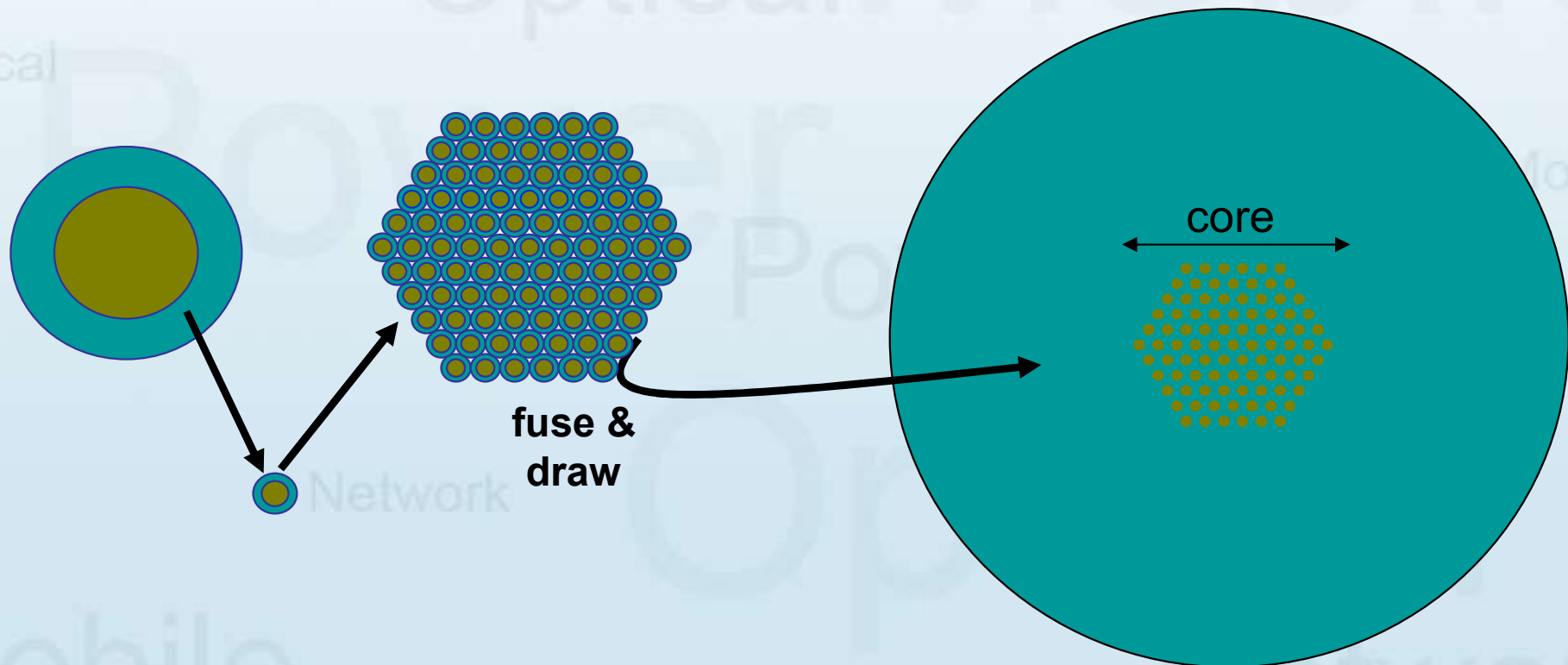
Coupling Coefficient (Experiment)



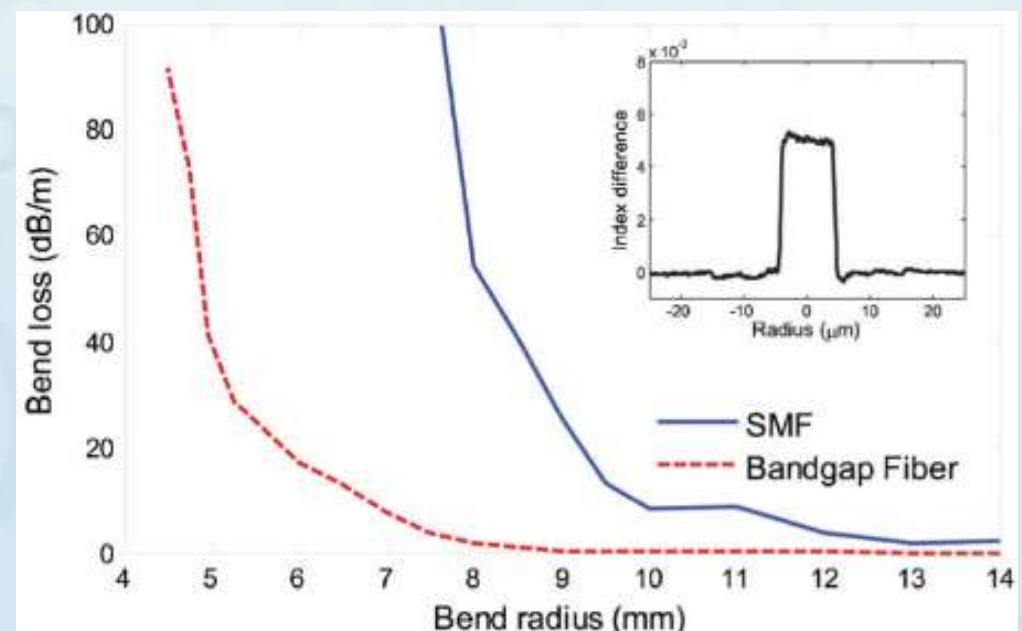
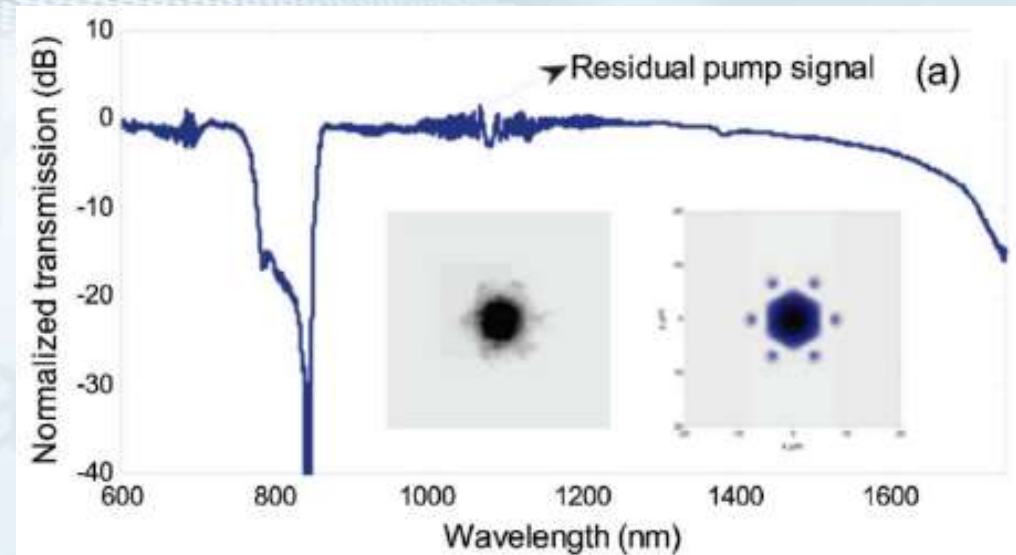
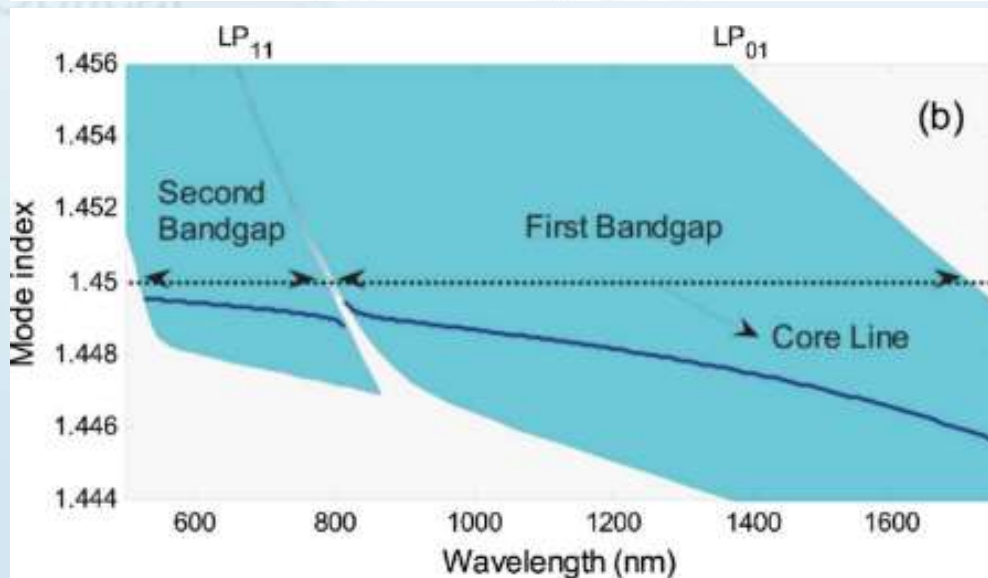
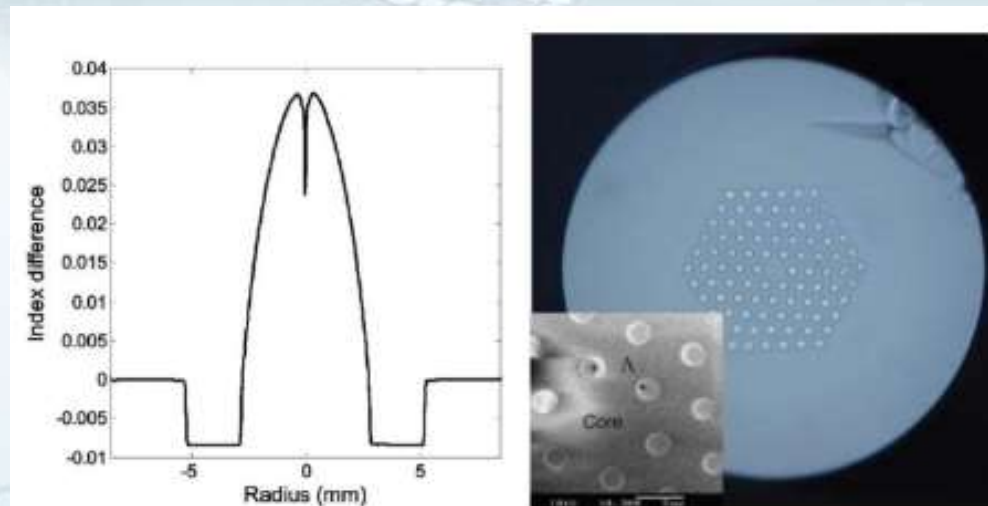
Nd:Yag ---- Neodymium Yttrium Aluminium Garnet Laser

Fabrication of Silica based NCF

- The background material is pure silica.
- The doped rods were etched to obtain the appropriate core/cladding diameter ratio.
- The preform was drawn to cane and stacked in a hexagonal pattern.
- This stacked core was overcladded with a pure-silica thick jacket.
- Finally the whole structure was drawn to realize the NCF.



3. All-Solid Bandgap Fiber (ASPBGF)



G. Ren *et al.*, Opt Lett **32**, 1023 (2007).

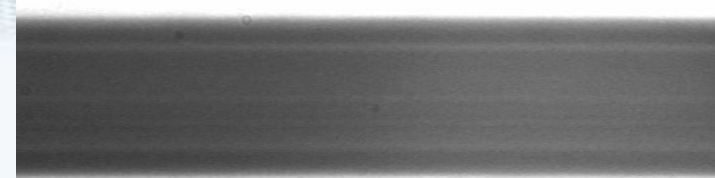
III. MOF based devices

1. Splicing on condensed photonic crystal fiber using CO₂ laser
2. MOF-based LPG by CO₂ laser
3. LPG in simplified hollow-core MOF and strain sensing application
4. Tunable filters based on LPG in MOF by microbending
5. Acoustic-optical LPG in simplified hollow-core MOF
6. Curvature sensor based on LPG in dual concentric-core fiber
7. FBG in Waveguide-array MOF
8. MOF-based Fabry–Pérot interferometer I
9. MOF-based Fabry–Pérot interferometer II
10. MOF-based MZI by core-offset and bubble
11. MZI based on LPG pair in MOF
12. MZI based on LPG and microtaper in ASPBGF
13. LPG cascaded to MOF-MZI for temperature and refractive index sensing
14. MOF-MZI based humidity sensor
15. MOF-MZI based ringdown strain sensor
16. MOF-MZI + FBG for simultaneous measurement to temperature & strain
17. MOF-MZI + LPG for simultaneous measurement to temperature & strain
18. Temperature Sensor by Using Selectively-Filled MOF Sagnac Interferometer
19. Twin-resonance-coupling in a selectively filled MOF and sensing applications

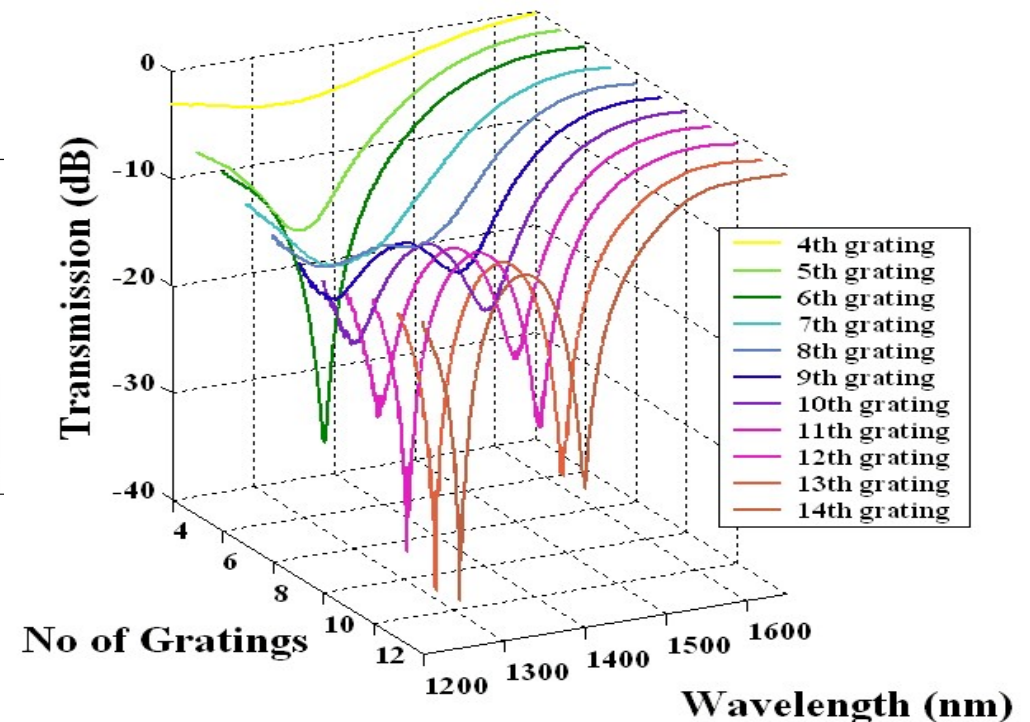
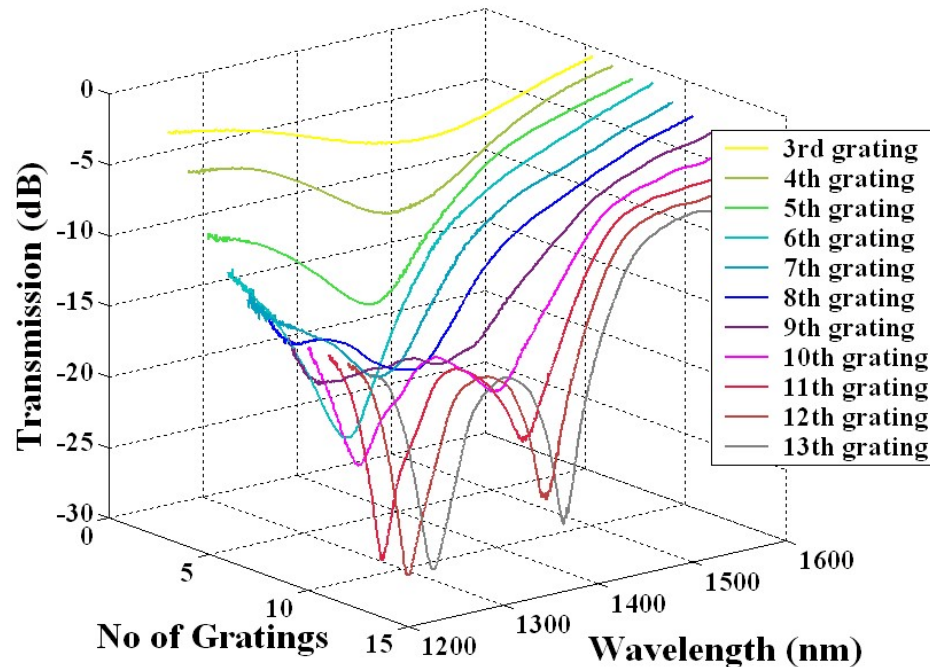
LPG's mechanisms: deformation and stress relaxation



**CO₂ deformation
(periodic tapering)**

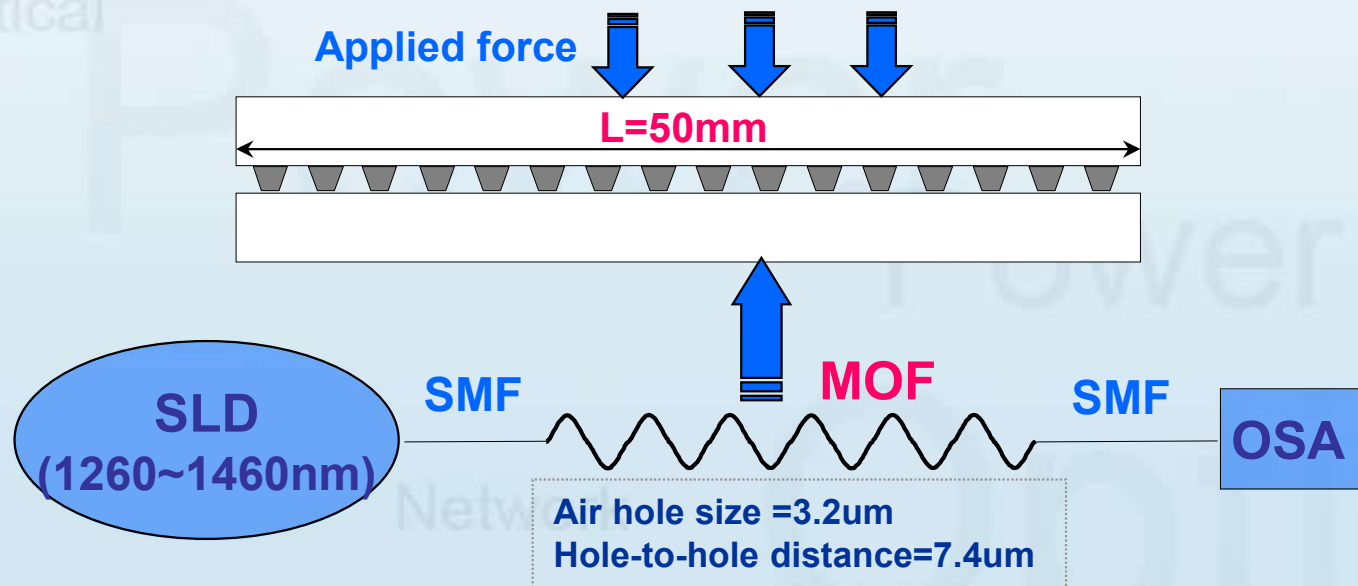
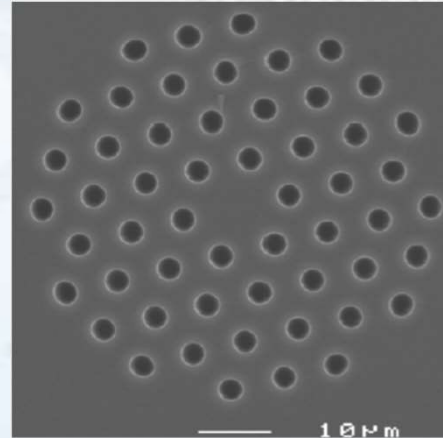
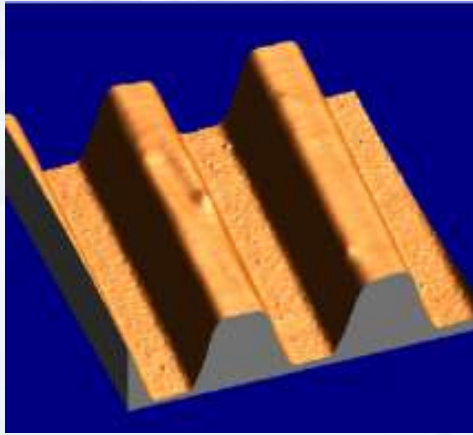


CO₂ stress relaxation

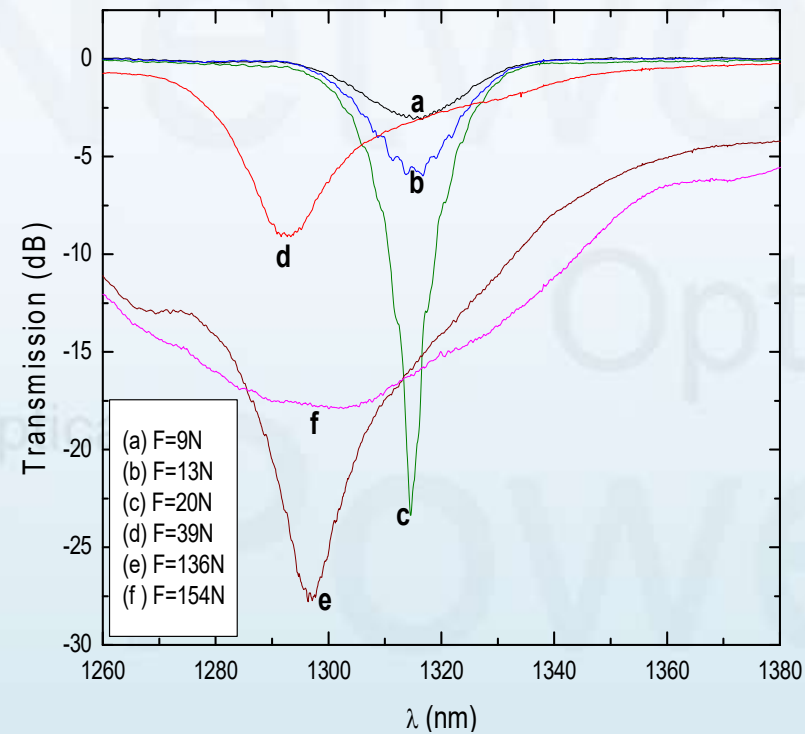


Evolution of resonance by increasing No. of gratings

4. MOF-LPG by microbending

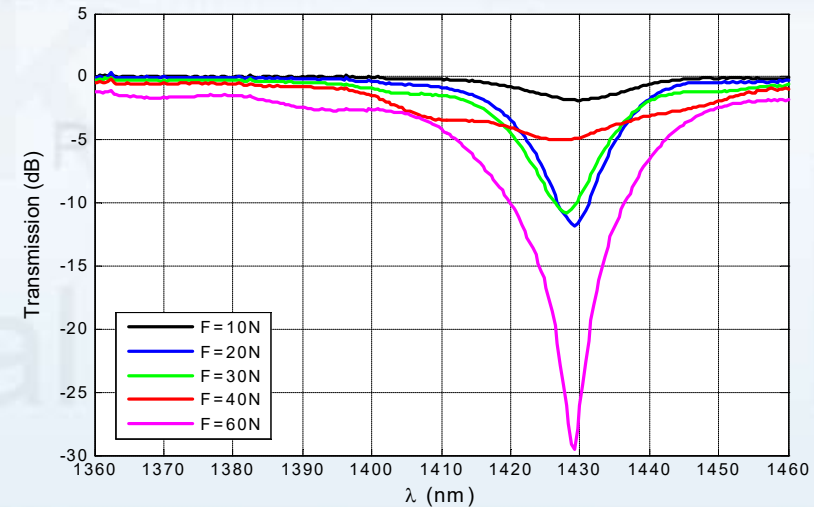


MOF-LPG by microbending

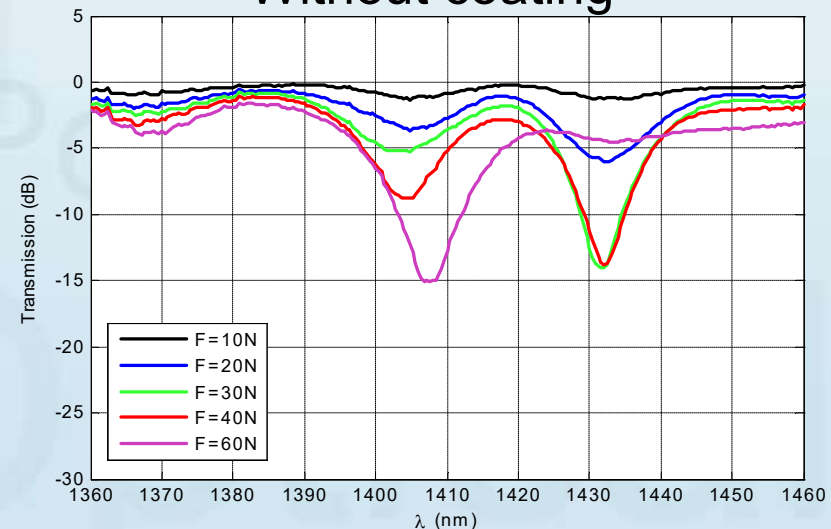


- Oscillation of transmission
- Blue shift of resonance wavelength

With coating

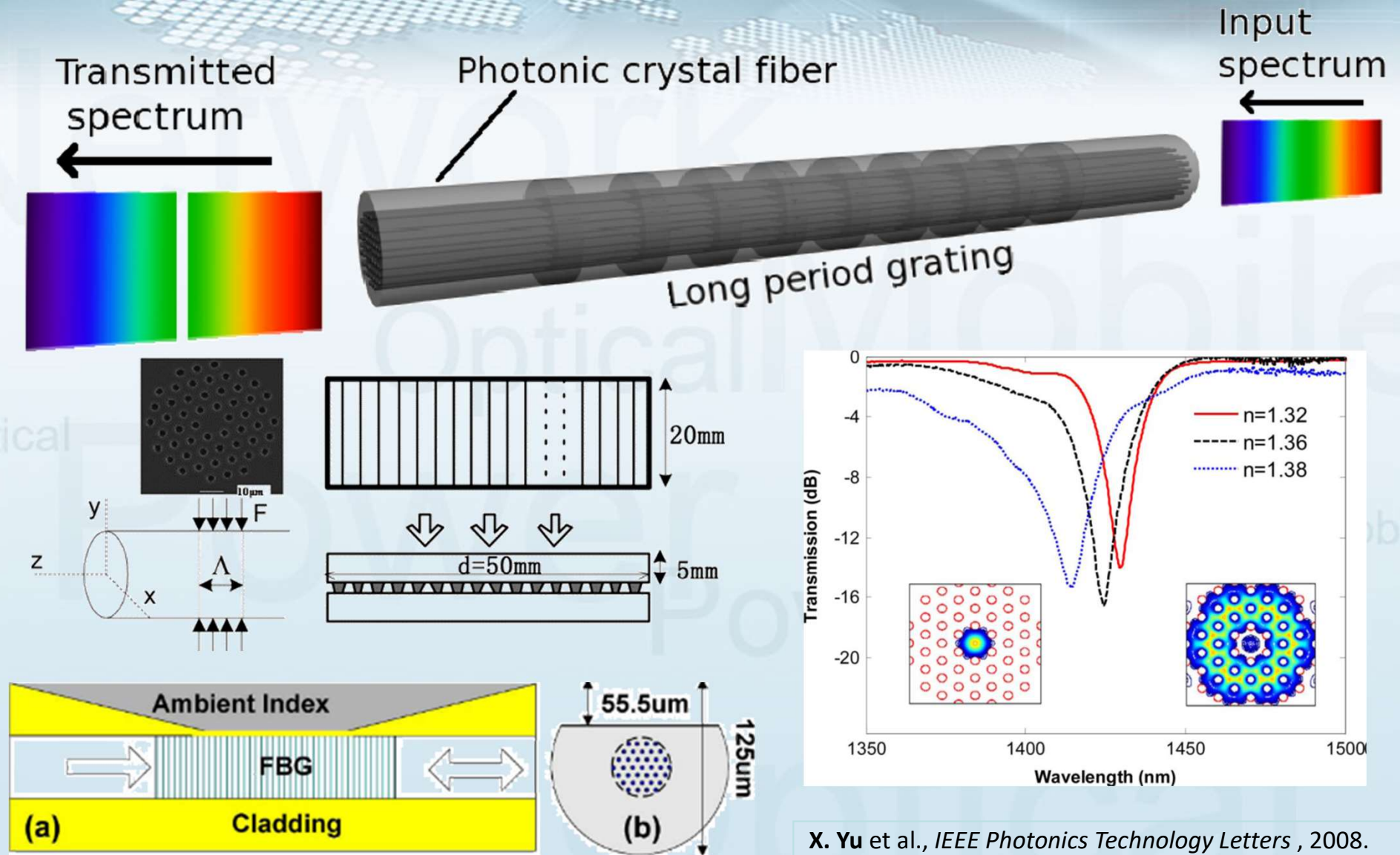


Without coating



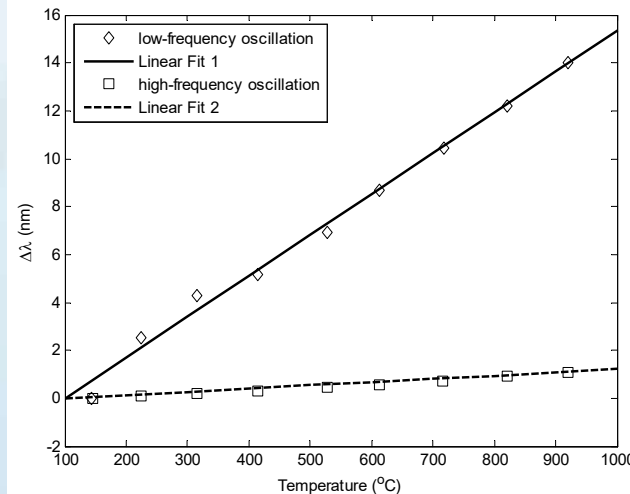
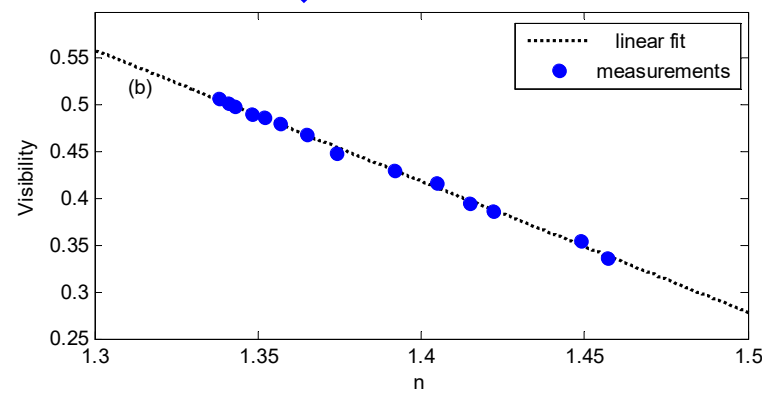
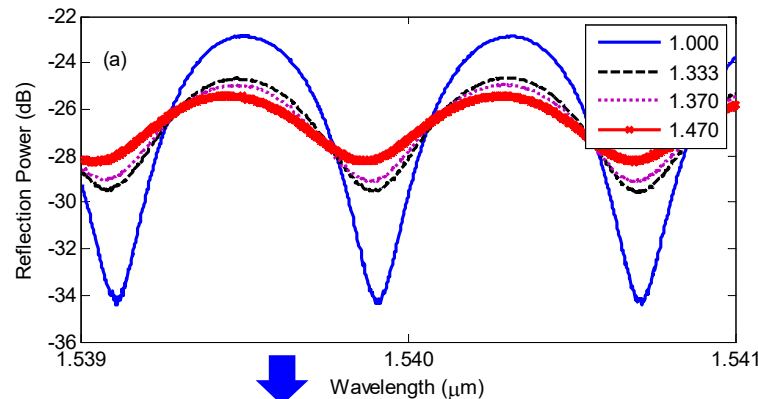
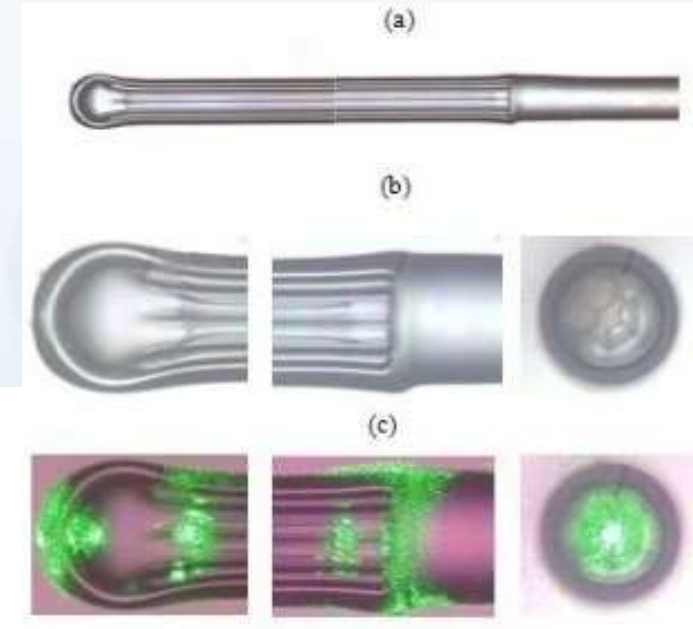
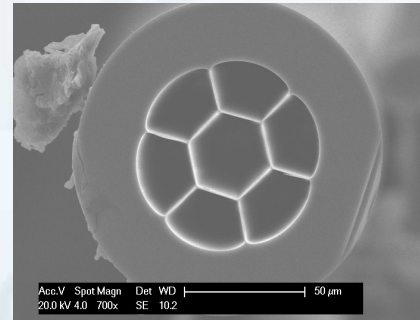
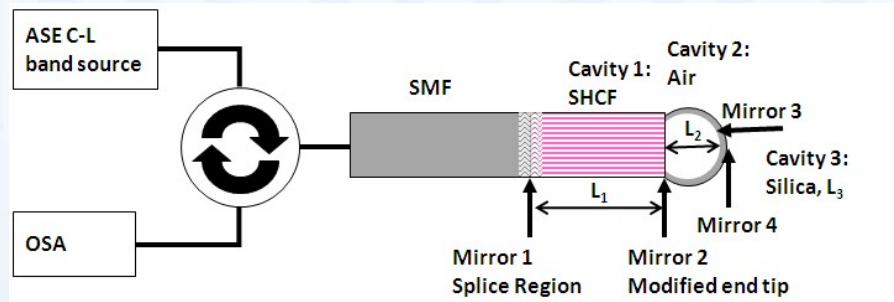
Effects of Acrylate coating

Tunable filters based on LPG in MOF



X. Yu et al., *IEEE Photonics Technology Letters*, 2008.
 X. Yu et al., *IEEE Journal of Lightwave Technology* 2009.

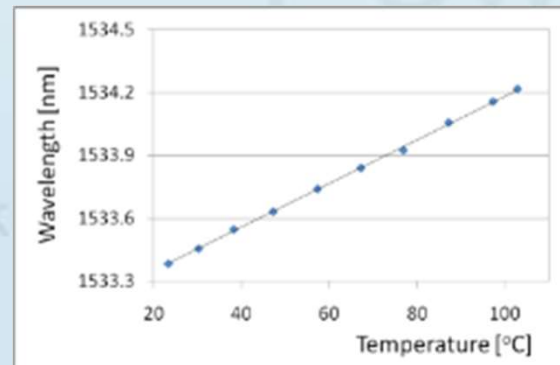
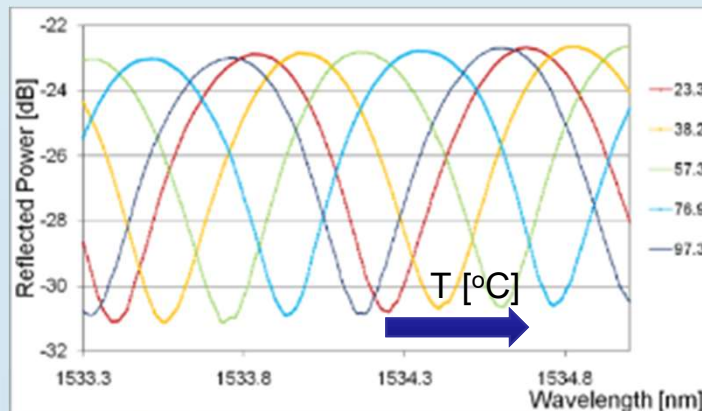
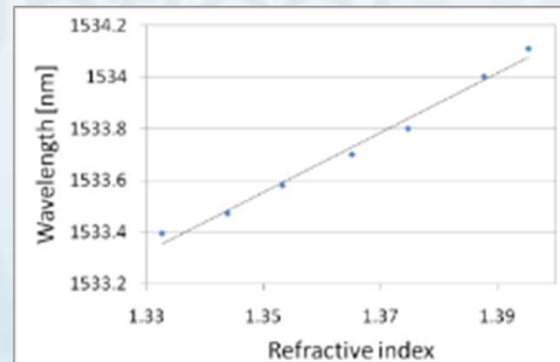
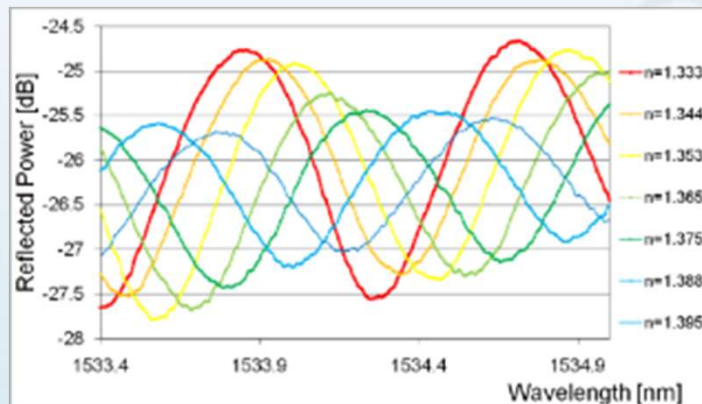
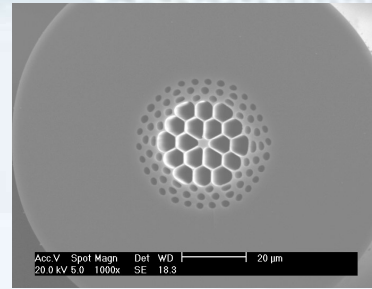
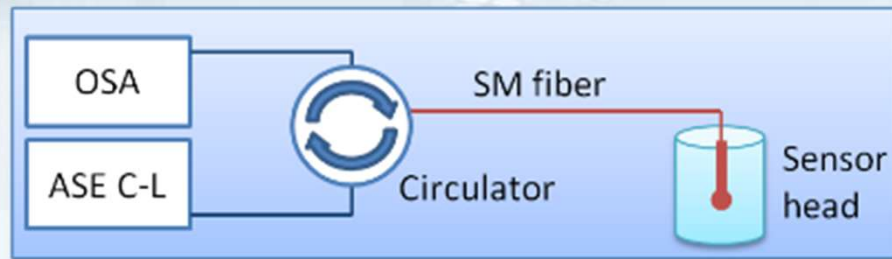
8. Fabry–Pérot Refractometer Based on a Simplified Hollow-Core Fiber with a Hollow Silica Sphere Tip



◀ Low temperature sensitivity of about 1.3 pm/°C for high frequency modulation fringes; high temperature sensitivity of about 17.064 pm/°C for low frequency modulation fringes.

◀ 6.2×10^{-4} refractive index (RI) resolution, when considering the OSA's visibility resolution of 0.01 dB.

9. MOF-based Fabry–Pérot interferometer

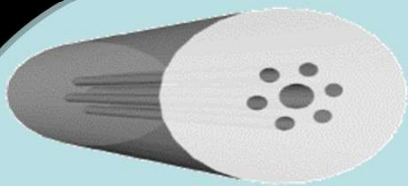


↑ Air bubble formed in the splice region causes core mode attenuation, and cladding mode excitation resulting in better interference pattern.

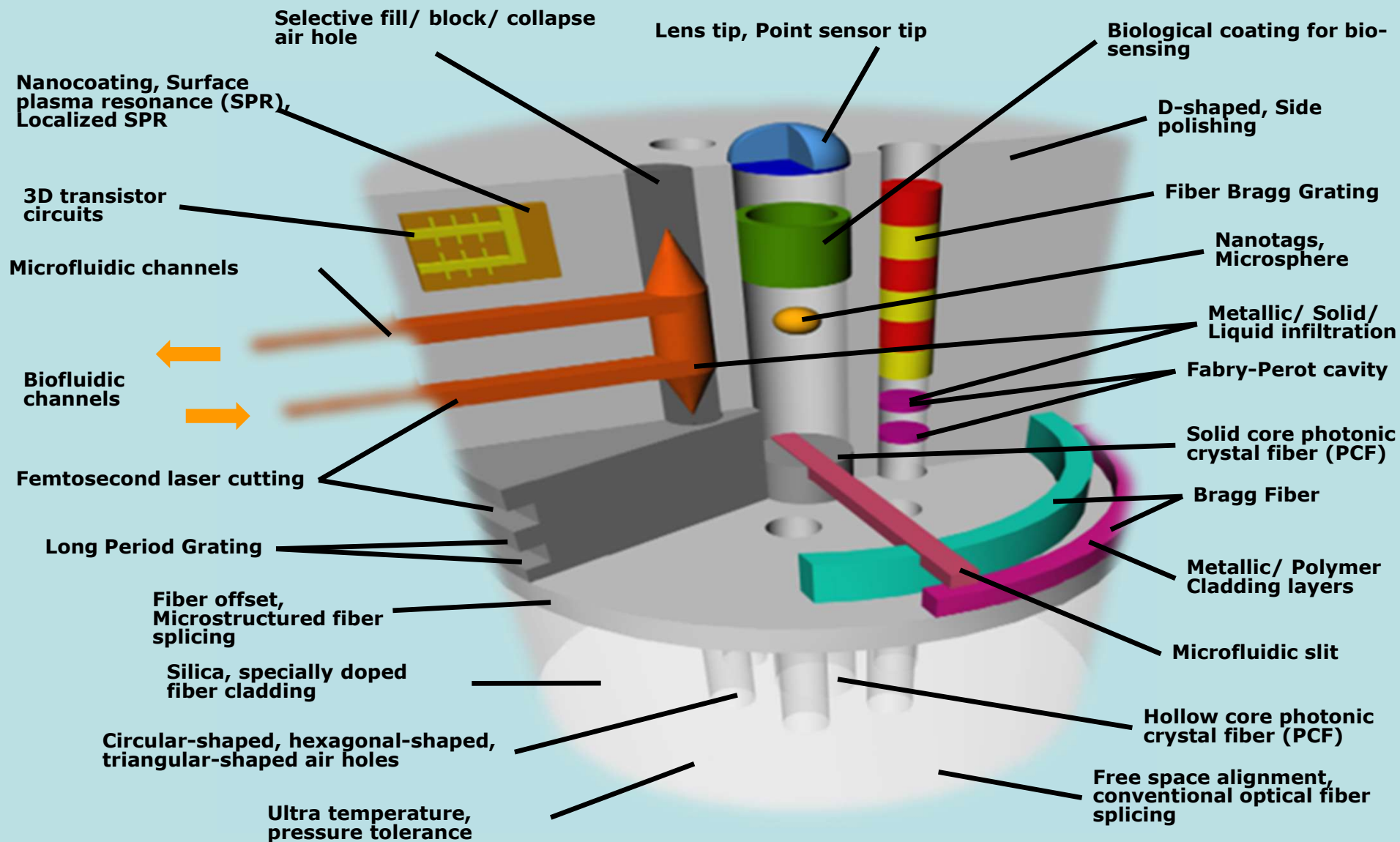
← Sensor has resolution in the order of 8.7×10^{-4} over the refractive index range of 1.33~1.40; and low temperature sensitivity in the order of 10 pm/°C in the range of 20~100 [°C].

Conclusions

Fiber sensors can facilitate distributed sensing, we can multiplex large number of sensors within a building e.g. 100. Long range remote sensing is possible > 50 km in range. High accuracy - fiber-optic gyroscope can exhibit stability of 1 part in 10^{18} . Moreover fiber is explosion proof, small size & light weight. Due to it's self-referencing nature, no recalibration/re-initialization will be needed. Useful technology for intelligent building sensing applications. e.g. Temperature, Electric field, Stress / Strain, Pressure, Acceleration, Birefringence and Polarization



Fibre Sensors



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